Corso di Laurea Magistrale in Matematica

Minimal Lagrangian orbits of Hamiltonian actions over KE and KRS

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Abstract

Hamiltonian actions of a Lie group on a symplectic manifold admit a special map, namely a *moment map*, from the manifold to the dual of the Lie algebra of the acting group. This map has surprisingly useful properties and becomes a tool to study geometric properties of the manifold. Among its valuable uses, we are concerned with its application to the study of Lagrangian orbits; in general, Lagrangian submanifolds represent a rich source of interest in geometry, partially motivated by the applications in mirror symmetry. We first give an explicit formula for the moment map, when the manifold acted on is a special type of symplectic manifold, i.e. Kähler Ricci Soliton. Then we study the minimality of Lagrangian orbits when the acting Lie group is semisimple. It turns out that, under certain hypotheses, there is only one Lagrangian orbit which is moreover minimal.

Finally, enters Godot

Introduction

The subject of my thesis falls within the broader scope of actions of Lie groups on symplectic and Kähler manifolds. When a Lie group G acts on a symplectic manifold M, it is possible to define a *moment map* $\mu : M \to \mathfrak{g}^*$ associated with the action; it can play a significant role in studying the geometric properties of the manifold itself. This map is defined on the manifold M and takes values in the dual of the Lie algebra of the acting group, and it is employed in mathematical literature in various ways.

Atiyah was the first to investigate convexity properties of the image of M via the moment map, in the case where the acting group G is abelian and the manifold M is compact. In particular, he proved that the image of the moment map is given by the convex hull of the image of the fixed points of the group G. Denoting $\mathfrak{t}$ as the Lie algebra of the maximal torus $T \subseteq G$ and $\mathfrak{t}^*_+$ as an appropriate subspace of $\mathfrak{t}^*$ called the *positive Weyl chamber*, we define $\Delta(M) = \mu(M) \cap \mathfrak{t}^*_+$. In the non-abelian case, Guillemin and Sternberg in [12] for M Kähler, and Kirwan more generally in [15] for M symplectic, showed that $\Delta(M)$ is a convex polytope known as the *moment polytope*. Delzant proved that, under suitable assumptions on the action, two manifolds admitting the same moment polytope are G-equivariantly symplectomorphic; interested readers are referred to [7] and [19] for further details. This highlights how $\Delta(M)$ should contain information about the action and how geometric properties of the orbits in the manifold can be reconstructed from it. To identify special points in $\Delta(M)$, the square norm of the moment map $||\mu||^2$ proves to be an important tool. Kirwan in [14] studied its properties. It becomes natural to understand which orbits in the manifold correspond to certain critical points of $||\mu||^2$.

Complex orbits correspond to the maximum point of $||\mu||^2$ (see [10]); in the case where G is semisimple, a Lagrangian orbit corresponds to the minimum point of $||\mu||^2$. It becomes interesting to study the minimality of the Lagrangian orbit. Pacini and subsequently Gori and Bedulli, under less restrictive assumptions on the action, proved that if the manifold M has a Kähler Einstein (KE) metric, then the Lagrangian orbit is minimal. In this thesis, we study the same problem in the case where the manifold M admits a Kähler Ricci Soliton (KRS) metric.

Combining the work [8] by Futaki with the solitonic equation, we find an expression of the moment map in the KRS case; moreover, under suitable assumptions, the result of Pacini, Gori, and Bedulli has been generalized in

the case where the group G is semisimple and the manifold is Fano, proving that the Lagrangian orbit, when it does exist, is also minimal. In the case where the metric is a KRS, the minimality of the Lagrangian orbit implies that it is critical for a Bott Morse function.
Most of the thesis is dedicated to the introduction and study of the concepts that are necessary, in the final chapter, for the analysis of the problem of the minimality of Lagrangian orbits.

As we need a basic understanding on how actions of Lie groups do work, in Chapter 1 we provide a soft introduction to their properties. By following [1], we fix notations and provide basic facts and definitions. We illustrate the role of the exponential map in the correspondence between Lie groups and Lie algebras and its relations with the adjoint representation and its differential. Moreover, the coadjoint representation is introduced due to its meaningful use for the moment map. Later in the chapter, the *Killing form* will bring about important definitions and concepts in this theory, such as *semisemplicity* (of Lie groups) and the decomposition of Lie algebras into *simple* ones. Finally, we talk about Lie group actions. In particular, a naive introduction to the nature of *orbits* takes place.

In Chapter 2, we explore the language of symplectic topology. We use two different sources here: [21] and [27]. After the linear setting is outlined, we focus on the main characters of the thesis: symplectic manifolds. Notions of symplectic form, Lagrangian submanifold, Hamiltonian function and others are introduced, until the way is paved for symplectic group actions. At this turning point, Hamiltonian actions are defined by mean of the moment map, and a detailed description of this last tool is supplied. Existence and uniqueness of moment maps are not trivial in general and we briefly discuss this matter.

By the end of the thesis we will be dealing with manifolds with underlying complex structure, hence now we want to interweave the complex and the symplectic structure together. We begin the Chapter 3 by briefly reviewing the concepts of (almost) complex structures over manifolds. Almost complex structures, usually denoted by J, are tensors that lead to the definition of many holomorphic tools on the manifold. When we consider an additional symplectic structure on (almost) complex manifolds, we do so that the symplectic form, the almost complex structure and the Riemannian metric are mutually related in a certain way, so that we are allowed to speak of *Kähler manifolds*. After a brief review of tensors and computation in local coordinates, we delve into Hodge theory and introduce various differential

operators, in order to build the groundings up for the subsequent chapters. *Harmonic* forms are considered and employed in the relevant theorems due to Hodge and Dolbeault. Finally, we define the first Chern class.
Throughout this chapter, we used as main sources [8], [21], [18], [23].

In Chapter 4, we introduce *Kähler Einstein* and *Kähler Ricci Soliton manifolds*. Their structures have been intensively studied in differential and complex geometry by many authors. We refer to [5], [29] and [30] for more in-depth sources. After the preliminary considerations, we consider an explicit formula due to Futaki in [8] for the moment map, when a compact (path) connected Lie group G acts on a compact connected Kähler manifold M. Then, we explicitly compute this formula in the case of M being a Kähler Ricci Soliton. A *canonical* moment map is already known and widely used in literature when the manifold is Kähler Einstein, and we check that this last moment map is indeed a particular case of the newly computed moment map. In this last part, we refer to [25] either.

Finally, in the last Chapter we are concerned with Lagrangian orbits. In [14], F. Kirwan studied in depth properties of the moment map and its relations with equivariant cohomology. In particular, a detailed analysis of geometric properties of the norm squared moment map is given. We begin by developing briefly this theory, historically relevant in Symplectic Geometry. At this point, we investigate Lagrangian orbits, by analyzing in depth some steps taken from [9]. Here, the topology of Lagrangian orbits is characterised under certain hypothesis. Once we have delved into the theory of Lagrangian orbits, we turn our attention to the understanding of their minimality. When does it happen that a Lagrangian orbit is also minimal? Which further hypotheses do we need to require in order for these orbits to become minimal? In [9], the case of Kähler Einstein manifolds is treated and we focus on its description. Finally, we are able to generalize one of these results to any *Fano manifold*, proving that, under some certain assumptions, whenever the acting Lie group is semisimple, there is only one Lagrangian orbit and it is minimal.
We conclude this work with a few remarks and possible perspectives.

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Chapter No. 1

A gentle introduction to Lie groups

In this chapter we assume the basic definitions of Lie groups and we take care of introducing the more sophisticated machinery that will be employed in the subsequent chapters. We follow [1]. Whenever G is a Lie group, we will denote its Lie algebra by $\mathfrak{g} := Lie(G)$.

1.1 The basics

We will indicate the *left* and *right translations* respectively as $L_g : G \to G, \quad x \mapsto gx$ and $R_g : G \to G, \quad x \mapsto xg$.
There is a strong relation between (connected) Lie subgroups and Lie subalgebras:

Theorem 1.1.1. *Let G be a Lie group with Lie algebra $\mathfrak{g}$, and $\mathfrak{h}$ be a Lie subalgebra of $\mathfrak{g}$. There exists a unique connected Lie subgroup $H \subset G$ with Lie algebra $\mathfrak{h}$.*

We are going to need the following theorem in order to define the exponential map, which is an extremely useful tool in differential geometry.

Theorem 1.1.2. *Let G_1 and G_2 be Lie groups, and $\vartheta : \mathfrak{g}_1 \to \mathfrak{g}_2$ be a Lie algebra homomorphism. There exists an open neighborhood V of e_1 and a smooth map $\varphi : V \to G_2$ that is a local homomorphism with $d\varphi_{e_1} = \vartheta$. In addition, if G_1 is connected and simply-connected, there exists a unique Lie group homomorphism $\varphi : G_1 \to G_2$ with $d\varphi_{e_1} = \vartheta$.*

Corollary 1.1.3. *If G_1 and G_2 are connected and simply-connected, and $\vartheta : \mathfrak{g}_1 \to \mathfrak{g}_2$ is an isomorphism, there exists a unique Lie group isomorphism*

$\varphi : G_1 \to G_2$ with $d\varphi_{e_1} = \vartheta$. In other words, if G_1 and G_2 are as above, then G_1 and G_2 are isomorphic if and only if $\mathfrak{g}_1$ and $\mathfrak{g}_2$ are isomorphic.

Let G be a Lie group and $\mathfrak{g}$ its Lie algebra. We recall that a Lie group homomorphism $\phi : \mathbb{R} \to G$ is called a 1-parameter subgroup of G. Let $X \in \mathfrak{g}$, and consider the Lie algebra homomorphism $\theta : \mathbb{R} \to \mathfrak{g}$, $\theta(t) := tX$. From Theorems 1.1.1 and 1.1.2, there is a unique 1-parameter subgroup $\lambda_X : \mathbb{R} \to G$ such that $\lambda_X(0) = e$.
It is remarkable that λ_X is an integral curve of the left-invariant vector field X passing through e:

$$\lambda_X(t) = \frac{d}{ds}\lambda_X(t+s)\Big|_{s=0} = \frac{dL_{\lambda_X(t)}\lambda_X(0)}{dt} = X(\lambda_X(t)).$$

Now we can clearly define the (Lie) **exponential map** of G

$$\exp : \mathfrak{g} \to G, \quad \exp(X) := \lambda_X(1),$$

where λ_X is the unique 1-parameter subgroup of G such that $\lambda_X(0) = X$. Many properties are listed in [1] which turn out to be interesting and somewhat generalize the concept of matrix exponential: this section mainly aims at fixing some notation and build the grounding mechanisms.

Let G be any group and V a vector space. We recall that a (linear) **representation** of G on V is a group homomorphism $\phi : G \to \mathrm{Aut}(V)$, where $\mathrm{Aut}(V)$ is the group of vector space isomorphisms of V.
Consider the action of G on itself by conjugation, i.e.,

$$a : G \times G \to G, \quad a(g,x) := a_g(x) = gxg^{-1}.$$

The representation

$$\mathrm{Ad} : G \to \mathrm{Aut}(\mathfrak{g}), \quad \mathrm{Ad}(g) := d(a_g)_e = d(L_g)_{g^{-1}} \circ d(R_g^{-1})_e,$$

is called the **adjoint representation** (or **adjoint action**) of G.
It follows that

$$\mathrm{Ad}(g)X = \frac{d}{dt}\left(g\exp(tX)g^{-1}\right)\Big|_{t=0}.$$

The differential of the adjoint representation Ad is denoted by ad and given by

$$\mathrm{ad} : \mathfrak{g} \to \mathrm{End}(\mathfrak{g}), \quad \mathrm{ad}(X)Y := d\mathrm{Ad}(e)(X)(Y).$$

It follows from the above definition that

$$\mathrm{ad}(X)Y = \frac{d}{dt}\left(\mathrm{Ad}(\exp(tX))Y\right)\Big|_{t=0}.$$

One may check that if $X, Y \in \mathfrak{g}$, then

$$\mathrm{ad}(X)Y = [X, Y].$$

Now, let $\langle \cdot, \cdot \rangle$ be the natural pairing between $\mathfrak{g}^*$ and $\mathfrak{g}$:

$$\langle \cdot, \cdot \rangle : \mathfrak{g}^* \times \mathfrak{g} \to \mathbb{R}, \quad (\xi, X) \mapsto \langle \xi, X \rangle = \xi(X).$$

Given $\xi \in \mathfrak{g}^*$, we define $\mathrm{Ad}^*(g)\xi$ by

$$\langle \mathrm{Ad}^*(g)\xi, X \rangle = \langle \xi, \mathrm{Ad}(g^{-1})X \rangle, \tag{1.1}$$

for any $X \in \mathfrak{g}$.
The collection of maps $\mathrm{Ad}^*(g)$ forms the **coadjoint representation** (or **coadjoint action**) of G on $\mathfrak{g}^*$:

$$\mathrm{Ad}^* : G \to \mathrm{Aut}(\mathfrak{g}^*), \quad g \mapsto \mathrm{Ad}^*(g).$$

We took this definition from [27]. Now we go on with [1].

The following result provides a characterization of connected abelian Lie groups.

Theorem 1.1.4. *Let G be a connected n-dimensional abelian Lie group. Then G is isomorphic to $T^k \times \mathbb{R}^{n-k}$, where $T^k = S^1 \times \ldots \times S^1$ is a k-torus. In particular, an abelian connected and compact Lie group is isomorphic to a torus.*

We end this section with few more important definitions. The *center of a Lie group G* is the subgroup given by

$$Z(G) := \{g \in G : ghg^{-1} = h, \text{ for all } h \in G\},$$

and the *center of a Lie algebra $\mathfrak{g}$* is defined as

$$Z(\mathfrak{g}) := \{X \in \mathfrak{g} : [X, Y] = 0, \text{ for all } Y \in \mathfrak{g}\}.$$

The following result relates the centers of a Lie group and its Lie algebra.

Lemma 1.1.5. *(i)* $Z(G) = \ker \mathrm{Ad}$*;*

(ii) $Z(G)$ *is a normal Lie subgroup of* G*;*

(iii) $Z(\mathfrak{g}) = \ker \mathrm{ad}$*;*

(iv) $Z(\mathfrak{g})$ *is the Lie algebra of* $Z(G)$*.*

1.2 The Killing form

Let G be a Lie group, and let $X, Y \in \mathfrak{g}$. The **Killing form** of $\mathfrak{g}$ (also called the Killing form of G) is the symmetric bilinear form

$$B(X, Y) := \mathrm{tr}(\mathrm{ad}(X)\mathrm{ad}(Y)).$$

If B is non-degenerate, then $\mathfrak{g}$ is said to be **semisimple**. A Lie group G is said to be semisimple if its Lie algebra $\mathfrak{g}$ is semisimple. We are going to need a different characterization of semisimplicity, which will needed in the coming chapters.

Lemma 1.2.1. *The Killing form is* Ad*-invariant, that is, for all* $X, Y \in \mathfrak{g}$ *and* $g \in G$*, it holds that* $B(\mathrm{Ad}(g)X, \mathrm{Ad}(g)Y) = B(X, Y)$*.*

Even though we don't strictly need it in the present thesis, the following result has its own degree of importance:

Theorem 1.2.2. *Let* G *be an* n*-dimensional semisimple connected Lie group. Then* G *is compact if and only if its Killing form* B *is negative-definite.*

If an ideal $\mathfrak{h}$ has no ideals other than the trivial ones $\{0\}$ and $\mathfrak{h}$, it is called *simple*. Following the usual convention, by *simple Lie algebras* we mean Lie subalgebras that are non-commutative simple ideals.
Given a Lie algebra $\mathfrak{g}$, consider the decreasing sequence of ideals

$$\mathfrak{g}^{(1)} := [\mathfrak{g}, \mathfrak{g}], \quad \mathfrak{g}^{(2)} := [\mathfrak{g}^{(1)}, \mathfrak{g}^{(1)}], \ldots, \mathfrak{g}^{(k)} := [\mathfrak{g}^{(k-1)}, \mathfrak{g}^{(k-1)}], \ldots$$

If there exists a positive integer m such that $\mathfrak{g}^{(m)} = \{0\}$, then $\mathfrak{g}$ is said to be **solvable**. One can prove that every Lie algebra admits a maximal solvable ideal τ, called its **radical**.
We can now provide a theorem giving equivalent definitions of semisimple Lie algebras.

Theorem 1.2.3. *Let $\mathfrak{g}$ be a Lie algebra with Killing form B. Then the following are equivalent:*

(i) $\mathfrak{g} = \mathfrak{g}_1 \oplus \ldots \oplus \mathfrak{g}_n$ *is the direct sum of simple Lie algebras* $\mathfrak{g}_i$ *(i.e., non-commutative simple ideals);*

(ii) $\mathfrak{g}$ *has trivial radical* $\tau = \{0\}$*;*

(iii) $\mathfrak{g}$ *has no commutative ideal other than* $\{0\}$*;*

(iv) B *is non-degenerate, i.e.,* $\mathfrak{g}$ *is semisimple.*

It is important to notice that if the Lie algebra $\mathfrak{g}$ is the direct sum of non-commutative simple ideals $\mathfrak{g} = \mathfrak{g}_1 \oplus \ldots \oplus \mathfrak{g}_n$, then $[\mathfrak{g}, \mathfrak{g}] = \mathfrak{g}$. Indeed, on the one hand, $[\mathfrak{g}_i, \mathfrak{g}_j] = \{0\}$ if $i \neq j$, since $\mathfrak{g}_i$ and $\mathfrak{g}_j$ are ideals. On the other hand, $[\mathfrak{g}_i, \mathfrak{g}_i] = \mathfrak{g}_i$, since $\mathfrak{g}_i^{(1)} = [\mathfrak{g}_i, \mathfrak{g}_i]$ is an ideal and $\mathfrak{g}_i$ is a non.commutative simple ideal. We are going to use this fact whenever speaking about semisimple Lie groups, instead of referring to the associated Killing form being non-degenerate.

A Riemannian metric $\langle \cdot, \cdot \rangle$ on a Lie group G is **left-invariant** if L_g is an isometry for all $g \in G$, that is, if for all $g, h \in G$ and $X, Y \in T_hG$,

$$\langle d(L_g)_h X, d(L_g)_h Y \rangle_{gh} = \langle X, Y \rangle_h.$$

Similarly, **right-invariant** metrics are those for which the right translations R_g are isometries. Quite not surprisingly, a **bi-invariant** metric Q on a Lie group G is a Riemannian metric that is simultaneously left and right-invariant. It is not so difficult for Lie groups to have bi-invariant metrics.

Theorem 1.2.4. *Every compact Lie group admits a bi-invariant metric Q.*

One can say a great deal about the Riemannian structure of a Lie group by mean of its bi-invariant metric, but this discussion exceeds the goal of the thesis. We are actually interested in the following lemma.

Lemma 1.2.5. *Let $\mathfrak{g}$ be a Lie algebra with a bi-invariant metric. Then $\mathfrak{g} = \tilde{\mathfrak{g}} \oplus Z(\mathfrak{g})$ is a direct sum of ideals, where $\tilde{\mathfrak{g}}$ is semisimple. In particular, $[\tilde{\mathfrak{g}}, \tilde{\mathfrak{g}}] = \tilde{\mathfrak{g}}$.*

The last result of this chapter is actually one of the most interesting result stated on the regard of Lie groups. We are going to need it in Chapter 5, when dealing with the squared norm of the moment map.

Theorem 1.2.6 (Maximal Torus Theorem)**.** *Let G be a connected, compact Lie group.*

1. *There exists a maximal torus T in G;*
2. *Any two maximal tori in G are conjugate;*
3. *Every element of G is contained in a maximal torus;*
4. *For each bi-invariant metric Q on G, the orbits of the conjugation action intersect every maximal torus Q-orthogonally.*

1.3 Lie group actions

Here we want to provide elementary definitions and results about Lie groups acting over smooth manifolds. Let G be a Lie group and M be a smooth manifold. Suppose that G acts on M (on the left). This means that there exists a smooth map $\psi : G \times M \to M$ satisfying

- $\psi(e, x) = x$ for all $x \in M$;
- $\psi(g_1, \psi(g_2, x)) = \psi(g_1 g_2, x)$ for all $g_1, g_2 \in G$ and $x \in M$.

For notation sake's we also define the auxiliary maps

$$\psi^g : M \to M \text{ and } \psi_x : G \to M$$

by

$$\psi^g : x \mapsto \psi(g, x) \text{ and } \psi_x : g \mapsto \psi(g, x).$$

When we want to point out the action of an element of $g \in G$ on some $x \in M$, we will write down equivalently $\psi^g(x), \psi(g, x)$ or $g \cdot x$, whenever it does not create confusion.

Definition 1.3.1. Let $\psi : G \times M \to M$ be a left action, and let $x \in M$. The subgroup

$$G_x := \{g \in G : \psi(g, x) = x\} \subset G$$

is called the **isotropy group** or **stabilizer** of $x \in M$, and

$$G(x) := \{\psi(g, x) : g \in G\} \subset M$$

is called the **orbit** of $x \in M$. If $G(x) = \{x\}$, then x is called a **fixed point** of the action.

The key idea of an action is that each $g \in G$ determines a transformation of M, namely $\psi^g : M \to M$. Since our actions are assumed smooth, then ψ^g are diffeomorphisms, and hence $\psi^G := \{\psi^g : g \in G\}$ can be identified with a subgroup of the diffeomorphism group $\mathrm{Diff}(M)$. An orbit $G(x)$ consists of all possible images $\psi^g(x)$ for $g \in G$, and the isotropy group G_x consists of all $g \in G$ that fix $x = \psi^g(x)$. If M is endowed with a Riemannian metric g, an action on (M, g) is said to be **isometric**, or *by isometries*, if ψ^g is an isometry of (M, g) for all $g \in G$. In this case, the metric g is said to be *G-invariant*, and ψ^G can be identified with a subgroup of $\mathrm{Iso}(M, g)$.

Definition 1.3.2. Consider two (left) G-actions $\psi_1 : G \times M_1 \to M_1$ and $\psi_2 : G \times M_2 \to M_2$. A map $f : M_1 \to M_2$ is called **G-equivariant** if $\psi_2(g, f(x)) = f(\psi_1(g, x))$ for all $x \in M_1$ and $g \in G$.

Definition 1.3.3. An action $\psi : G \times M \to M$ is **proper** if the map $G \times M \to M \times M$, $(g, x) \mapsto (\psi(g, x), x)$, is proper.

Definition 1.3.4. In the previous notation, for each $X \in \mathfrak{g}$, we define the **action field** $\widehat{X}$ by setting

$$\widehat{X}_p := \frac{d}{dt}\bigg|_{t=0} \psi(\exp(tX), p) \in T_pM. \tag{1.2}$$

The correspondence $X \mapsto \widehat{X}$ is called **infinitesimal action**.

One can check that any action field $\widehat{X}$ belongs in the $C^\infty(M)$-module of vector fields $\mathfrak{X}(M)$ and that moreover its flow is given by $\varphi^t_{\widehat{X}} = \psi^{\exp(tX)}$.
The following proposition is essential in that it allows us to understand better orbits of actions. In particular, it answers questions naturally arising whenever dealing with subsets of manifolds: are these objects submanifolds? How are they immersed? When does the immersion become an embedding? All of these doubts are solved in the following.

Proposition 1.3.5. *Let $\psi : G \times M \to M$ be a left action and define $\tilde{\psi}_x : G/H \to M$ by $\tilde{\psi}_x \circ \rho = \psi_x$, where $H = G_x$ is the isotropy at $x \in M$ and $\rho : G \to G/H$ is the quotient map. Then $\tilde{\psi}_x$ is a G-equivariant injective immersion, with image $G(x)$. In particular, $G(x)$ is an immersed submanifold of M whose tangent space at $x \in M$ is $T_xG(x) = d(\psi_x)_e(g)$. In addition, if the action is proper, then $\tilde{\psi}_x$ is an embedding and $G(x)$ is an embedded submanifold of M.*

$$\begin{tikzcd} G \arrow[d, "\rho"'] \arrow[dr, "\psi_x"] & \\ G/H \arrow[r, "\tilde{\psi}_x"'] & M \end{tikzcd}$$

From above we understand that the tangent space $T_xG(x)$ to an orbit is the image of the linear map $d(\psi_x)_e : \mathfrak{g} \to T_xM$. Denote by $\mathfrak{g}_x$ the Lie algebra of the isotropy group G_x, and let $\mathfrak{m}_x$ be such that $\mathfrak{g} = \mathfrak{g}_x \oplus \mathfrak{m}_x$ as vector spaces. Since $\ker d(\psi_x)_e = \mathfrak{g}_x$, we have that $\mathfrak{m}_x$ can be identified with $T_xG(x)$. More precisely, there is a natural identification

$$\mathfrak{m}_x \xrightarrow{\sim} \widehat{X}_x \in T_xG(x)$$

where $\widehat{X}$ is the action field on M induced by $X \in \mathfrak{m}_x$.
We have established that orbits represent very interesting subsets of manifolds acted on by Lie groups and, in case they are submanifolds, we have understood the nature of their tangent space. Now we want to distinguish between them.

Definition 1.3.6. Let $\psi : G \times M \to M$ be a proper action. Then $G(x)$ is a **principal orbit** if there exists a neighborhood V of $x \in M$ such that for each $y \in V$, $G_x \subset G_{\psi(g,y)}$ for some $g \in G$.

Informally speaking, principal orbits are those having the smallest isotropy group among the nearby orbits.

Definition 1.3.7. Let $\psi : G \times M \to M$ be a proper action.

1. The orbit $G(x)$ has a **larger orbit type** than $G(y)$ if there is a G-equivariant map $\varphi : G(x) \to G(y)$, or, equivalently, if there is $g \in G$ such that $G_x \subset G_{\psi(g,y)}$.

2. The orbits $G(x)$ and $G(y)$ have the **same orbit type** if there is a G-equivariant diffeomorphism $\varphi : G(x) \to G(y)$, or, equivalently, if there is $g \in G$ such that $G_x = G_{\psi(g,y)}$.

3. An orbit $G(x)$ is said to be **regular** if the dimension of $G(x)$ coincides with the dimension of principal orbits.

4. A nonprincipal regular orbit is called **exceptional**.

5. A nonregular orbit is called **singular**.

It is not difficult to see that having the same orbit type is an equivalence relation between orbits and hence between points of M: x and y in M are in the same equivalence class if and only if their respective orbits $G(x)$ and $G(y)$ are of the same type.

Definition 1.3.8. A **stratification** of a manifold M is a partition of M by embedded submanifolds $\{M_i\}_{i\in I}$ of M, called **strata**, such that:

1. The partition is locally finite, i.e., each compact subset of M only intersects a finite number of strata;

2. For each $i \in I$, there exists a subset $I_i \subset I \setminus \{i\}$ such that the closure of M_i is
$$\overline{M_i} = M_i \cup \bigcup_{j\in I_i} M_j;$$

3. $\dim M_j < \dim M_i$ for all $j \in I_i$.

There exists a preferred stratification of a manifold allowing for a proper action. Let

$$M_x := \{y \in M| \quad y \text{ has the same orbit type of } x\}.$$

Theorem 1.3.9. *Let $\psi : G\times M \to M$ be a proper action. The partition of M into connected components of the orbit types M_x determines a stratification.*

Chapter No. 2

Symplectic Topology in a nutshell

Symplectic topology has its roots deep in classical mechanics and possesses many connections to miscellaneous areas of mathematics. It represents the perfect geometric environment for Hamiltonian formalism, which arises from a Legendre transformation and leads to the notion of *canonical transformation*.
We don't delve into the historical transition from classic to modern formalism, for which a nice resource is the first chapter of [21], but we briefly recall the basics of symplectic topology and thus dive into more advanced concepts, such as moment maps and their properties. Again, in this first part we will be adherent to [21].

2.1 Symplectic Vector Spaces

As it is common habit, we need to address the easier linear case before and only after will we treat symplectic manifolds.
The standard example of symplectic vector space is the Euclidean space $\mathbb{R}^{2n}$ endowed with the skew-symmetric form

$$\omega_0 = \sum_{j=1}^{n} dx_j \wedge dy_j,$$

where $(x_1, \ldots, x_n, y_1, \ldots, y_n)$ are coordinates on $\mathbb{R}^{2n}$. More generally, we define a **symplectic vector space** to be a pair (V, ω), where V is a real finite-dimensional vector space and ω a non-degenerate skew-symmetric bilinear form $\omega : V \times V \to \mathbb{R}$. Let us recall that it means the following:

skew-symmetry: for all $u, v \in V$,

$$\omega(u, v) = -\omega(v, u);$$

nondegeneracy: for any $v \in V$,

$$\omega(v, u) = 0 \text{ for every } u \in V \text{ implies } v = 0.$$

A central role in this theory is played by a peculiar sort of homomorphims, i.e. those that preserve the symplectic structure of these symplectic vector spaces. We define an isomorphism $\Psi : V \to V$ of a symplectic vector space (V, ω) to be a **linear symplectomorphism** whenever $\Psi^*\omega = \omega$, where

$$(\Psi^*\omega)(u, v) := \omega(\Psi u, \Psi v) \text{ for all } u, v \in V.$$

Now we introduce some standard terminology.
The **symplectic complement** of a linear subspace $W \subset V$ is defined as the subspace

$$W^\omega = \{v \in V | \quad \omega(v, w) = 0 \text{ for all } w \in W\}.$$

A subspace W can be

- **isotropic** if $W \subset W^\omega$;
- **coisotropic** if $W^\omega \subset W$;
- **symplectic** if $W \cap W^\omega = \{0\}$;
- **Lagrangian** if $W = W^\omega$.

The following result implies that all symplectic vector spaces of the same dimension are linearly symplectomorphic (i.e. there exists a symplectomorphism between them).

Theorem 2.1.1. *Let (V, ω) be a symplectic vector space of dimension $2n$. Then there exists a basis $u_1, \ldots, u_n, v_1, \ldots, v_n$ such that*

$$\omega(u_j, u_k) = \omega(v_j, v_k) = 0 \text{ and } \omega(u_j, v_k) = \delta_{jk}.$$

*Such a basis is called a **symplectic basis**. Moreover, there exists a vector space isomorphism $\Psi : \mathbb{R}^{2n} \to V$ such that $\Psi^\star\omega = \omega_0$.*

Actually one can prove a slightly more general result concerning vector spaces endowed with a skew-symmetric bilinear form and, by exhibiting an explicit basis, derive the previous statement and conclude that symplectic vector spaces need to be even-dimensional.
Also, from the theorem above one can deduce the following

Corollary 2.1.2. *Let V be a $2n$-dimensional vector space and let ω be a skew-symmetric bilinear form on V. Then ω is non-degenerate if and only if its n-fold exterior product is non-zero, i.e. $0 \neq \omega^n = \omega \wedge \cdots \wedge \omega$.*

2.2 Symplectic Manifolds

Unless otherwise specified, we will assume that M is a differential manifold without boundary. We will refer to M as being closed whenever it is compact and without boundary.
A **symplectic structure** on M is a non-degenerate closed 2-form $\omega \in \Omega^2(M)$. Non-degenerate means that (T_qM, ω_q) is a symplectic vector space for each $q \in M$. Thus the manifold is $2n$- dimensional and, by Corollary 2.1.2, the n-fold product of ω never vanishes. This implies that M is oriented.
Any form such as ω is a **symplectic form**. It is worth noticing that such forms satisfy two conditions. The first is purely algebraic and implies the existence of a canonical bundle isomorphism

$$TM \to T^*M, \quad X \mapsto \iota_X\omega = \omega(X, \cdot), \tag{2.1}$$

while the second is closedness and is geometric.
A **symplectomorphism** of a symplectic manifold M is a diffeomorphism $\psi \in \mathrm{Diff}(M)$ such that

$$\psi^*\omega = \omega.$$

Being ω non-degenerate, (2.1) is a bijection and hence there is a one-to-one correspondence between vector fields and 1-forms given by

$$\mathfrak{X}(M) \to \Omega^1(M), \quad X \mapsto \iota_X\omega.$$

A vector field $X \in \mathfrak{X}(M)$ is called **symplectic** if $\iota_X\omega$ is closed.
Now consider instead a smooth function $H : M \to \mathbb{R}$. The vector field defined by

$$\iota_{X_H}\omega = dH$$

is called the **Hamiltonian vector field** associated to the **Hamiltonian function** H.
When M is closed, X_H generates a smooth 1-parameter group of diffeomorphisms φ_H^t satisfying

$$\frac{d}{dt}\varphi_H^t = X_H(\varphi_H^t), \quad \varphi_H^0 = \mathrm{id}.$$

This is called the **Hamiltonian flow** associated to H.
It is noticeable that the vector field X_H is tangent to level sets $H = const$ of H, indeed one has

$$dH(X_H) = \iota_{X_H}\omega(X_H) = \omega(X_H, X_H) = 0.$$

More generally, it happens that a smooth function $F \in C^\infty(M)$ is constant along the orbits of the flow of H if and only if the **Poisson Bracket**

$$\{F, H\} = \omega(X_F, X, H) = dF(X_H)$$

vanishes. One can prove that this bracket endows $C^\infty(M)$ with a Lie algebra structure.

Remark 2.2.1. *Before going on with the theory apparatus, let us justify briefly why this framework is such a prolific environment for dynamical systems. We define a* ***Hamiltonian system*** *as a triple (M, ω, H), where (M, ω) is a symplectic manifold and $H \in C^\infty(M)$ is called the* ***Hamiltonian function****. As observed above, a function $f \in C^\infty(M)$ is constant along the integral curves of X_H if and only if $\{f, H\} = 0$. Such f is called an* ***integral of motion*** *(or a* ***first integral*** *or a* ***constant of motion****).*
In general, Hamiltonian systems do not admit integrals of motion which are independent of the Hamiltonian function. Functions $f_1, \ldots, f_n$ on M are called ***independent*** *if their differentials $d_pf_1, \ldots, d_pf_n$ are linearly independent at all points $p \in M$ in some open dense subset of M. Loosely speaking, a Hamiltonian system is completely integrable if it has as many commuting integrals of motion as possible. Commutativity is intended with respect to the Poisson Bracket. Notice that, if $f_1, \ldots, f_n$ are commuting integrals of motion for a Hamiltonian system (M, ω, H), at each $p \in M$ their Hamiltonian vector fields generate an isotropic subspace of T_pM:*

$$\omega(X_{f_i}, X_{f_j}) = \{f_i, f_j\} = 0.$$

If moreover $f_1, \ldots, f_n$ are independent, by symplectic algebra, the number n can be at most half the dimension of M. An Hamiltonian system (M, ω, H) is ***completely integrable*** *if if possesses $n = \frac{1}{2} dim(M)$ independent integrals of motion $f_1 = H, f_2, \ldots, f_n$, which pairwise commute with respect to the Poisson Bracket.*

The following lemma describes some interesting properties related to Hamiltonian flows.

Lemma 2.2.2. *Let (M, ω) be a symplectic manifold.*

1. *Whenever defined, the Hamiltonian flow φ_H^t is a symplectomorphism which is tangent to the level surfaces of H.*

2. *For every Hamiltonian function $H : M \to \mathbb{R}$ and every symplectomorphism $\psi : M \to M$ we have $X_{H \circ \psi} = \psi^* X_H$.*

3. *The Lie bracket of two Hamiltonian vector fields X_F and X_G is the Hamiltonian vector field $[X_F, X_G] = X_{\{F,G\}}$.*

Proof. Notice that the first statement follows easily from Cartan's magic formula:

$$\frac{d}{dt}(\varphi_H^t)^*\omega = \varphi_H^t(\mathcal{L}_{X_H}\omega) = (\varphi_H^t)^*(\iota_{X_H} d\omega + d\iota_{X_H}\omega) =$$

$$(\varphi_H^t)^*(d\iota_{X_H}\omega) = (\varphi_H^t)^*(d^2 H) = 0.$$

Since $\varphi_H^0 = \mathrm{id}$, it follows that $(\varphi_H^t)^*\omega = \omega$ at each time.
The second statement is the identity

$$\iota_{X_{H\circ\psi}}\omega = d(H \circ \psi) = \psi^* dH = \psi^* \iota_{X_H}\omega = \iota_{\psi^* X_H}\omega.$$

In order to prove the third statement, since φ_F^t is a symplectomorphism, consider

$$[X_F, X_G] = -\frac{d}{dt}\bigg|_{t=0} (\varphi_F^t)^* X_G = -\frac{d}{dt}\bigg|_{t=0} X_{G\circ\varphi_F^t}.$$

Hence

$$\iota_{[X_F,X_G]}\omega = -\frac{d}{dt}\bigg|_{t=0} d(G \circ \varphi_F^t) = -d\frac{d}{dt}\bigg|_{t=0} G \circ \varphi_F^t =$$

$$-d(dG(X_F)) = -d\{G, F\} = d\{F, G\}.$$

□

In analogy with the definitions given in the linear setting, we say that a submanifold $Q \subset M$ is **Lagrangian** (or **symplectic**, **isotropic**, **coisotropic**) if for each $q \in Q$ the subspace T_qQ of (T_qM, ω_q) is Lagrangian (or symplectic, isotropic, coisotropic).
Finally, the following is the manifold-wise counterpart of Theorem (2.1.1):

Theorem 2.2.3 (Darboux)**.** *Every symplectic form ω on M is locally diffeomorphic to the standard form ω_0 on $\mathbb{R}^{2n}$.*

Notice that this theorem implies that symplectic manifolds, unlike Riemannian manifolds, admit no local invariant: indeed locally they are just indistinguishable.

2.3 Symplectic Group Actions

In this section we add some substance and begin dealing with actions of Lie groups over symplectic manifolds. Here we will blend material coming from [1], [21], [27].
Let us fix notation and consider a symplectic manifold (M, ω) acted on (the left) by a Lie group G.
For simplicity suppose in the following that G is connected and let $\mathfrak{g}$ be its Lie algebra. Then the exponential map $\exp : \mathfrak{g} \to G$ is well defined.
So far we haven't advanced any request dealing with the underlying symplectic structure of M. We require that the action preserves the symplectic structure.
If we define $\mathrm{Sympl}(M, \omega)$ to be the set of symplectomorphisms $\varphi : M \to M$, our hypotheses will be that

(M, ω) is a symplectic manifold,
G is a Lie group,
$\psi : G \to \mathrm{Sympl}(M, \omega)$ is a symplectic action, i.e. it preserves the symplectic structure of M.

The following definition is fundamental to this work:

Definition 2.3.1. The action ψ is a **Hamiltonian action** if there exists a map

$$\mu : M \to \mathfrak{g}^*$$

satisfying:

1. For each $\xi \in \mathfrak{g}$, let
   - $\mu^\xi : M \to \mathbb{R}$, $\mu^\xi(p) := \langle \mu(p), \xi \rangle$, be the component of μ along ξ;
   - $\widehat{\xi}$ the action field associated to $X \in \mathfrak{g}$.

   Thus
   $$d\mu^\xi = \iota_{\widehat{\xi}}\omega,$$
   which means that μ^ξ is a Hamiltonian function for $\widehat{\xi}$.

2. μ is *equivariant* with respect to the given function ψ of G on M and the coadjoint action Ad^*, defined in (1.1), of G on $\mathfrak{g}^*$:
   $$\mu \circ \psi^g = \mathrm{Ad}^*(g) \circ \mu,$$
   for all $g \in G$.

The map μ is called a **moment map** and the vector (M, ω, G, μ) is sometimes called **Hamiltonian G-space**.

Remark 2.3.2. *One may wonder, whenever an Hamiltonian action is endowed with two moment maps $\mu_1, \mu_2 : M \to \mathfrak{g}^*$, what relations interweave them together. Consider $\xi \in \mathfrak{g}$. Since both μ_1^ξ and μ_2^ξ are Hamiltonian functions for $\widehat{\xi}$, it follows that $\mu_1 - \mu_2$ is locally constant. Under the assumption that M is connected, we may say that it is indeed constant everywhere, which means that there exists $c = c(\xi) \in \mathfrak{g}^*$ such that*
$$\mu_1(p) = \mu_2(p) + c,$$
for any $p \in M$. Hence, for any $g \in G$,
$$\mathrm{Ad}^*(g)\mu_1 = \mathrm{Ad}^*(g)\mu_2 + \mathrm{Ad}^*(g)c = \mu_2 \circ \psi^g + \mathrm{Ad}^*(g)c.$$
Now we look at the left-hand side:
$$\mathrm{Ad}^*(g)\mu_1 = \mu_1 \circ \psi^g = (\mu_2 + c) \circ \psi^g = \mu_2 \circ \psi^g + c.$$
We conclude that
$$\mathrm{Ad}^*(g)c = c \in \mathfrak{g}^*,$$
i.e.
$$c \in Z(\mathfrak{g}^*) = \mathrm{Ann}([\mathfrak{g}, \mathfrak{g}]).$$
Hence, we obtained that different moment maps for the same action do coincide but of an element of $\mathrm{Ann}([\mathfrak{g}, \mathfrak{g}])$.

Remark 2.3.3 (Angular momentum)**.** *The name of this map is pretty evocative and can be justified by the following. Suppose we are dealing with a Hamiltonian system. If $H : M \to \mathbb{R}$ is a Hamiltonian function which is invariant under the action of G, then H is an integral for the flow of H_ξ for every $\xi \in \mathfrak{g}$. Hence $\{H, H_\xi\} = 0$ and thus the moment map is constant along the integral curves of the flow generated by H. The angular momentum in $\mathbb{R}^3$ is a good example of this, which is the origin of the term moment map. Consider the action of $G = SO(3)$ on $\mathbb{R}^6 \simeq \mathbb{R}^3 \times \mathbb{R}^3$ (endowed with the standard symplectic structure) by*

$$\psi_\Phi(x, y) := (\Phi x, \Phi y)$$

for $\Phi \in SO(3)$. One can check that the action is generated by functions

$$H_A(x, y) := \langle y, Ax \rangle$$

for $A = -A^t \in \mathfrak{so}(3) := \mathrm{Lie}(\mathrm{SO}(3))$. To examine the moment map, note that the Lie algebra $\mathfrak{so}(3)$ can be identified with $\mathbb{R}^3$ via the map

$$\mathbb{R}^3 \to \mathfrak{so}(3) : \xi \mapsto A_\xi := \begin{pmatrix} 0 & -\xi_3 & \xi_2 \\ \xi_3 & 0 & -\xi_1 \\ -\xi_2 & \xi_1 & 0 \end{pmatrix}.$$

Thus, $A_\xi x = \xi \times x$ for $x, \xi \in \mathbb{R}^3$, and

$$[A_\xi, A_\eta] = A_{\xi \times \eta}, \quad A_{\Phi\xi} = \Phi A_\xi \Phi^{-1}, \quad \mathrm{tr}(A_\xi^\top A_\eta) = 2\langle \xi, \eta \rangle.$$

The last identity implies that the standard inner product on $\mathbb{R}^3$ induces an invariant inner product on $\mathfrak{so}(3)$, so that it can be identified with its dual $\mathfrak{so}(3)^*$ via this inner product. With this notation, the Hamiltonian function H_{A_ξ} can be written in the form

$$H_{A_\xi}(x, y) = \langle y, \xi \times x \rangle = \langle x \times y, \xi \rangle,$$

and hence the moment map $\mu : \mathbb{R}^3 \times \mathbb{R}^3 \to \mathfrak{so}(3)$ is given by

$$\mu(x, y) = A_{x \times y}.$$

If we think of x as the position and y as the momentum coordinate, then $x \times y$ is the angular momentum. Now, if $H : \mathbb{R}^6 \to \mathbb{R}$ is any Hamiltonian function which depends only on $|x|$, $|y|$, and $\langle x, y \rangle$ (e.g. in the case of motion in a central force field), then the function H is invariant under the diagonal action of $\mathrm{SO}(3)$. Hence, in this case, the angular momentum determines three independent integrals of the motion.

Whenever G is path-connected, Hamiltonian actions can be equivalently defined in terms of a **comoment map**

$$\mu^* : \mathfrak{g} \to C^\infty(M)$$

with the two conditions rephrased as follows:

- $\mu^*(\xi) := \mu^\xi$ is a Hamiltonian function for the vector field $\widehat{\xi}$;
- μ^* is a Lie algebra homomorphism:

$$\mu^*[\xi, \zeta] = \{\mu^*(\xi), \mu^*(\zeta)\}.$$

The equivalence of these two different definitions is proven in the following

Lemma 2.3.4. *Let $G \to Sympl(M, \omega) : g \mapsto \psi^g$ be a symplectic action by a compact Lie group, let $\mathfrak{g} \to \mathfrak{X}(M) : \xi \to \widehat{\xi}$ be the infinitesimal action and let $\mathfrak{g} \to C^\infty(M) : \xi \to H_\xi$ be a linear map such that $\iota_{\widehat{\xi}}\omega = dH_\xi$ for each $\xi \in \mathfrak{g}$. Consider the following assertions:*

1. *the map $\xi \to H_\xi$ is G-equivariant, i.e.*

$$H_\xi \circ \psi^g = H_{Ad(g^{-1})\xi}$$

   for all $\xi \in \mathfrak{g}$ and $g \in G$;

2. *the map $\xi \to H_\xi$ is a Lie algebra homomorphism with respect to the Poisson structure on $C^\infty(M)$, i.e.*

$$\{H_\xi, H_\zeta\} = H_{[\xi,\zeta]}$$

   for all $\xi, \zeta \in \mathfrak{g}$.

Then 1. implies 2.. If G is path-connected, then 1. and 2. are equivalent.

Proof. We assume first that G is path connected and prove that 2. implies 1.. One can check that $\widehat{\mathrm{Ad}(g^{-1})\xi} = (\psi^g)^*\widehat{\xi}$ and hence $H_{\mathrm{Ad}(g^{-1})\xi}$ and $H_{\xi\circ\psi^g}$ generate the same Hamiltonian vector field. So their difference is constant and, by 2., this implies that

$$H_{\mathrm{Ad}(g^{-1})[\xi,\eta]} = \{H_{\mathrm{Ad}(g^{-1})\xi}, H_{\mathrm{Ad}(g^{-1})\eta}\} = \{H_\xi \circ \psi^g, H_\eta \circ \psi^g\} =$$

$$\{H_\xi, H_\eta\} \circ \psi^g = H_{[\xi,\eta]} \circ \psi^g$$

for $\xi, \eta \in \mathfrak{g}$ and $g \in G$. Now let $g_1 \in G$ and $\xi \in \mathfrak{g}$ be given. Since G is path connected there exists a smooth path $g : [0,1] \to G$ connecting $g(0) = e$ and $g(1) = g_1$. Define $\eta(t) := \dot{g}(t)g(t)^{-1} \in \mathfrak{g}$. Then

$$\frac{d}{dt}\psi^g = \widehat{\eta} \circ \psi^g, \quad \frac{d}{dt}\mathrm{Ad}(g^{-1})\xi = \mathrm{Ad}(g^{-1})[\xi, \eta]$$

and hence

$$\frac{d}{dt}(H_\xi \circ \psi^g - H_{\mathrm{Ad}(g^{-1})\xi}) = d(H_\xi \circ \psi^g)(\psi^g)^*\widehat{\eta} - H_{\mathrm{Ad}(g^{-1})[\xi,\eta]} =$$

$$\omega(X_{H_{\xi\circ\psi^g}}, (\psi^g)^*\widehat{\eta}) - H_{[\xi,\eta]} \circ \psi^g =$$

$$\omega(X_{H_{\xi\circ\psi^g}}, X_{H_\eta\circ\psi^g}) - \{H_\xi, H_\eta\} \circ \psi^g =$$

$$\{H_{\xi\circ\psi^g}, H_\eta \circ \psi^g\} - \{H_\xi, H_\eta\} \circ \psi^g = 0.$$

This implies $H_\xi \circ \psi^{g_1} = H_{\mathrm{Ad}_{g_1^{-1}}\xi}$.
In order to see that 1. implies 2. we simply differentiate the identity

$$H_\xi \circ \psi^{exp(t\eta)} = H_{exp(-t\eta)\xi exp(t\eta)}$$

at $t = 0$.

□

2.4 Further comments on moment maps

In this section we want to address few remarks on moment maps of Hamiltonian group actions. Since they come along with an aura of arbitrariness and mystery, we want to understand when these objects exist and, if so, are uniquely determined on the vector (M, ω, G). Our reference here is [27].
Suppose that G is path-connected, so that the existence of moment maps or comoment maps stands for the same condition. For notation's sake, let us define the real vector space of symplectic vector fields and Hamiltonian vector fields over M respectively as $\mathfrak{X}^{\mathrm{sympl}}(M)$ and $\mathfrak{X}^{\mathrm{ham}}(M)$. Thus it is clear that we have the following exact sequences

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathfrak{X}^{\mathrm{sympl}}(M) & \hookrightarrow & \mathfrak{X}^{\mathrm{ham}}(M) & \longrightarrow & H^1(M;\mathbb{R}) & \longrightarrow & 0 \\ & & & & \xi & \longmapsto & [\iota_\xi\omega] & & \end{array} \tag{2.2}$$

and

$$0 \longrightarrow \mathbb{R} \hookrightarrow C^\infty(M) \longrightarrow \mathfrak{X}^{\mathrm{ham}}(M) \longrightarrow 0$$

$$f \longmapsto \xi_f$$

Now, if $\psi : G \times M \to M$ is the action, let us consider the infinitesimal action $d\psi : \mathfrak{g} \to \mathfrak{X}^{\mathrm{sympl}}(M)$, defined by $X \mapsto \widehat{X}$. Notice that it is a Lie algebra homomorphism.
We can restate the condition of ψ being Hamiltonian by saying that it is indeed Hamiltonian if and only if there is a a Lie algebra homomorphism $\mu^* : \mathfrak{g} \to C^\infty(M)$ lifting $d\psi$. i.e. making the following diagram commute:

$$\begin{tikzcd} C^\infty(M) \arrow[rr] & & \mathfrak{X}^{\mathrm{ham}}(M) \\ & \mathfrak{g} \arrow[ul, "\mu^*"] \arrow[ur, "d\psi"'] & \end{tikzcd} \tag{2.3}$$

Obviously, μ^* is again the comoment map.
In order to question the existence and uniqueness of moment map, let us define first the *Lie algebra cohomology*. For any Lie algebra and in particular for $\mathfrak{g} = \mathrm{Lie}(G)$ we define the k-cochains on $\mathfrak{g}$ as

$$C^k := \Lambda^k \mathfrak{g}^*$$

and a linear operator $\delta : C^k \to C^{k+1}$ by

$$\delta c(X_0, \dots, X_k) := \sum_{i<j} (-1)^{i+j} c([X_i, X_j], X_0, \dots, \widehat{X}_i, \dots, \widehat{X}_j, \dots, X_k).$$

One can check that $\delta^2 = 0$.
The **Lie algebra cohomology groups** of $\mathfrak{g}$ are the cohomology groups of the complex $\{C^k, \delta : C^k \to C^{k+1}\}_{k \in \mathbb{N}}$:

$$H^k(\mathfrak{g}; \mathbb{R}) := \frac{\ker\delta : C^k \to C^{k+1}}{\mathrm{im}\delta : C^{k-1} \to C^k}.$$

There is a theorem that tells us how to compute such groups in certain conditions:

Theorem 2.4.1. *If $\mathfrak{g}$ is the Lie algebra of some compact and path connected Lie group G, then*

$$H^k(\mathfrak{g};\mathbb{R}) = H^k_{dR}(G;\mathbb{R}).$$

One may wonder what is the meaning of this new cohomology. In low degrees we are able to understand it pretty well:

- An element in $C^1 = \mathfrak{g}^*$ is a linear functional on $\mathfrak{g}$. If $c \in C^1$, then $\delta c(X_0, X_1) = -c([X_0, X_1])$. Recall that the **commutator ideal** of $\mathfrak{g}$ is

  $$[\mathfrak{g},\mathfrak{g}] := \{\text{linear combination of } [X,Y] \text{ for any } X, Y \in \mathfrak{g}\}.$$

  Since $\delta c = 0$ if and only if it vanishes on $[\mathfrak{g},\mathfrak{g}]$, we conclude that

  $$H^1(\mathfrak{g};\mathbb{R}) = \mathrm{Ann}([\mathfrak{g},\mathfrak{g}]).$$

- An element of C^2 is an alternating bilinear map $c : \mathfrak{g} \times \mathfrak{g} \to \mathbb{R}$. One has

  $$\delta c(X_0, X_1, X_2) = -c([X_0, X_1], X_2) + c([X_0, X_2], X_1) - c([X_1, X_2], X_0).$$

  If $c = \delta b$ for some $b \in C^1$, then

  $$c(X_0, X_1) = (\delta b)(X_0, X_1) = -b([X_0, X_1]).$$

The following theorem allows us to say something about existence of moment maps:

Theorem 2.4.2. *If $H^1(\mathfrak{g};\mathbb{R}) = H^2(\mathfrak{g};\mathbb{R}) = 0$, then any symplectic G-action is Hamiltonian.*

So, when is $H^1(\mathfrak{g};\mathbb{R}) = H^2(\mathfrak{g};\mathbb{R}) = 0$? Recall that when a Lie group G is semisimple, then $\mathfrak{g} = [\mathfrak{g},\mathfrak{g}]$.

Theorem 2.4.3 (Whitehead Lemmas)**.** *Let G be a compact Lie group. Then G is semisimple if and only if $H^1(\mathfrak{g};\mathbb{R}) = H^2(\mathfrak{g};\mathbb{R}) = 0$.*

It follows then that if G is semisimple, any symplectic action is Hamiltonian. Now, we want to question uniqueness of moment maps. Suppose that G is a compact path connected Lie group.

Theorem 2.4.4. *If $H^1(\mathfrak{g};\mathbb{R}) = 0$, then moment maps for Hamiltonian G-actions are unique.*

Remark 2.4.5. *Notice that G being semisimple implies the uniqueness of the moment map directly from Remark 2.3.2. Indeed one has that $Z(\mathfrak{g}) = 0$, so that in the previous notations $c = 0$ and $\mu_1 = \mu_2$.*

So far we have seen that the topology of the acting Lie group G can play a fair role in existence and uniqueness of Hamiltonian actions. Actually, the topology of the underlying manifold M sometimes has its own importance in this matter. Indeed, suppose for instance M to be simply connected, i.e. $\pi_1(M) = 0$. This implies that $H^1(M;\mathbb{R}) = 0$; by looking at (2.2) we understand that

$$\mathfrak{X}^{\mathrm{sympl}}(M) \simeq \mathfrak{X}^{\mathrm{ham}}(M).$$

We can that conclude that the lifting in (2.3) can be naturally worked out and the action has got to be Hamiltonian.

Chapter No. 3

Complex and Kähler Geometry

This chapter briefly develops the theory of almost complex manifolds and links it to that of complex manifolds. After some relations are outlined, we blend those structures together with the symplectic one and see which conditions one has to consider in order for them to be compatible. Lastly, we introduce some Hodge theory for a better understanding of Chapter 4. We also define the first Chern class for a very specific case.
Our main references here will be [18], [23] and [16].

3.1 Almost Complex Structures

A **complex structure** J on a finite dimensional vector space consists in a generalization of the multiplication by i of $\mathbb{C}^n$. Indeed it is defined to be an endomorphism $J : V \to V$ such that $J^2 = -1$, i.e. when applied twice it behaves as minus the identity map. When the vector space is $\mathbb{C}^n \simeq \mathbb{R}^{2n}$, one can choose

$$J_n = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}.$$

We could develop first the linear theory and then apply it to the manifold environment as we did before, but here we choose to go straight to the geometric setting and skip the algebraic one.

Let M be a smooth manifold.

Definition 3.1.1. An **almost complex structure** on a M is a tensor field J which is, at every point $x \in M$, an endomorphism of the tangent space T_xM satisfying $J^2 = -1$. A manifold endowed with such a structure is called **almost complex manifold**.

By developing at first this theory on real vector spaces, one can then show that almost complex manifolds are necessarily even-dimensional and orientable.
Often one endows manifolds with a **complex structure**, by choosing a holomorphic atlas over the manifold. It is easy to check in local coordinates that a complex structure gives rise naturally to an almost complex structure; one may wonder under which hypothesis an almost complex structure allows for a complex structure. We will see that it is not as much trivial as the converse, being the almost complexity a weaker condition than the complexity.

Lemma 3.1.2. *A map $f : \mathbb{C}^n \to \mathbb{C}^m$ preserves the almost complex structures of $\mathbb{C}^n$ and $\mathbb{C}^m$, i.e. $f_* \circ J_n = J_m \circ f_*$, if and only if f is holomorphic.*

In order to define an almost complex structure on a complex manifold M, we transfer the almost complex structure of $\mathbb{C}^n$ to M by means of chart. Then it follows from Lemma 3.1.2 that an almost complex structure can be defined on M independently of the choice of charts. Now we say that an almost complex structure J on a manifold M is called complex structure on M if M is an underlying smooth manifold of a complex manifold which induces J in the way just described.

Remark 3.1.3. *It is relevant noticing that Lemma 3.1.2 allows us to characterize holomorphic maps between (almost) complex manifolds as those preserving the (almost) complex structure. Indeed locally this condition is equivalent to the Cauchy-Riemann equations being satisfied.*

Now we want to take a closer look to what new is given on the manifold by the almost complex structure J.
We would like to diagonalise the endomorphism J, but in order to do so we have to complexify each tangent space: define

$$TM^{\mathbb{C}} := TM \otimes \mathbb{C}.$$

We extend all real endomorphisms and differential operators from TM to $TM^{\mathbb{C}}$ by $\mathbb{C}$-linearity. Let $T^{1,0}M$ (resp. $T^{0,1}M$) denote the eigenbundles of $TM^{\mathbb{C}}$ corresponding to the eingenvalue i (resp. $-i$) of J. Then one can check that

$$T^{1,0}M := \{X - iJX | X \in TM\}, \quad T^{0,1}M := \{X + iJX | X \in TM\}$$

and $TM^{\mathbb{C}} = T^{1,0}M \oplus T^{0,1}M$.
In the same spirit we introduce the complexified exterior bundle

$$\Lambda^*_{\mathbb{C}} := \Lambda^* \otimes \mathbb{C}.$$

Sections of this bundle can be considered either as complex valued forms or as formal sums $\omega + i\tau$, where both ω and τ are usual real forms on M.
We define the following sub-bundles of $\Lambda^1_{\mathbb{C}}$:

$$\Lambda^{1,0}M := \{\xi \in \Lambda^1_{\mathbb{C}} | \xi(Z) = 0 \text{ for each } Z \in T^{0,1}M\}$$

and

$$\Lambda^{0,1}M := \{\xi \in \Lambda^1_{\mathbb{C}} | \xi(Z) = 0 \text{ for each } Z \in T^{1,0}M\}.$$

One can prove that

$$\Lambda^{1,0}M\{\omega - i\omega \circ J | \omega \in \Lambda^1 M\}, \quad \Lambda^{0,1}M\{\omega + i\omega \circ J | \omega \in \Lambda^1 M\}$$

and $\Lambda^1_{\mathbb{C}} = \Lambda^{1,0}M \oplus \Lambda^{0,1}M$.
More generally, one can prove that

$$\Lambda^k_{\mathbb{C}} \simeq \bigoplus_{p+q=k} \Lambda^{p,q}M,$$

where $\Lambda^{p,q}M := \Lambda^{p,0}M \otimes \Lambda^{0,q}M$, $\Lambda^{p,0}M := \bigotimes^p \Lambda^{1,0}M$, and $\Lambda^{0,q}M := \bigotimes^q \Lambda^{0,1}M$.
One should also notice that ω is a section of $\Lambda^{p,q}M$ if and only if it vanishes whenever applied to $p+1$ vectors in $T^{1,0}M$ or to $q+1$ vectors in $T^{0,1}M$.

Notation As we already did before, sometimes we are slightly imprecise in notation, for instance when we refer to vector fields that are taken in the tangent bundle TM instead of its module of sections $\mathfrak{X}(M) := \Gamma(TM)$. This fact is common even in literature; sometimes we will be adherent to it, whilst some others we will adopt the proper notation. Let us fix the notation. We will use $\Omega^k(M) := \Gamma(\Lambda^k M)$ and $\Omega^{p,q}(M) := \Gamma(\Lambda^{p,q}M)$; whenever referring to section of complexified bundles, we will use $\mathfrak{X}^{\mathbb{C}}(M) := \Gamma(TM^{\mathbb{C}})$ and $\Omega^k(M, \mathbb{C}) := \Gamma(\Lambda^k_{\mathbb{C}}M)$.

To any almost complex structure J one can associate a $(2,1)$-tensor N^J called the **Nijenhuis tensor**, satisfying

$$N^J(X,Y) := [X,Y] + J[JX,Y] + J[X,JY] - [JX,JY]$$

for $X, Y \in \mathfrak{X}(M)$. By use of this new tensor, we can add further characterization of almost complex structures:

Proposition 3.1.4. *Let J be an almost complex structure on M^{2n}. Then the following statements are equivalent:*

(a) *J is a complex structure;*

(b) *$T^{0,1}M$ is integrable;*

(c) *$d\Omega^{1,0}(M) \subset \Omega^{2,0}(M) \oplus \Omega^{1,1}(M)$;*

(d) *$d\Omega^{p,q}(M) \subset \Omega^{p+1,q}(M) \oplus \Omega^{p,q+1}(M)$;*

(e) *$N^J = 0$.*

Let us go on and consider a complex manifold (M, J), where J satisfies any of the above conditions. Say $\dim_{\mathbb{C}}(M) = m$. We can characterize homolorphic functions in the following way.

Lemma 3.1.5. *Let $f : M \to \mathbb{C}$ be a smooth complex-valued function on M. The following assertions are equivalent:*

(a) *f is holomorphic;*

(b) *$Z(f) = 0$ for all $Z \in T^{0,1}M$;*

(c) *df is a form of type $(1, 0)$.*

By using Proposition 3.1.4, for every fixed pair (p, q) we define the differential operators $\partial : \Omega^{p,q}(M) \to \Omega^{p+1,q}(M)$ and $\bar{\partial} : \Omega^{p,q}(M) \to \Omega^{p,q+1}(M)$ by $d = \partial + \bar{\partial}$.
Notice that, since

$$0 = d^2 = (\partial + \bar{\partial})^2 = \partial^2 + \bar{\partial}^2 + (\partial\bar{\partial} + \bar{\partial}\partial)$$

and the three operators take values in different sub-bundles, it follows

$$\partial^2 = 0, \quad \bar{\partial}^2 = 0, \quad \partial\bar{\partial} + \bar{\partial}\partial = 0.$$

Definition 3.1.6. A vector field $Z \in \Gamma(T^{1,0}M)$ is called **holomorphic** if $Z(f)$ is holomorphic for every locally defined holomorphic function f. A p-form ω of type $(p, 0)$ is called holomorphic if $\bar{\partial}\omega = 0$. A real vector field X is called **real holomorphic** if $X - iJX$ is a holomorphic vector field.

We can get a better understanding of real holomorphic vector fields:

Lemma 3.1.7. *Let X be a real vector field on (M, J) complex manifold. The following statements are equivalent:*

(a) *X is real holomorphic;*

(b) $\mathcal{L}_X J = 0$;

(c) *the flow of X consists of holomorphic transformations of M.*

Notice that we will call holomorphic a map $f : (M_1, J_1) \to (M_2, J_2)$ between two complex manifold whenever its differential commutes with the complex structures at each point: $f_* \circ J_1 = J_2 \circ f_*$. This is done in analogy with the previously given definitions.

The following result is very much important in complex geometry:

Proposition 3.1.8 (The local $i\partial\bar{\partial}$-Lemma)**.** *Let $\omega \in \Lambda^{1,1} \cap \Lambda^2 M$ be a real 2-form of type $(1,1)$ on a complex manifold M. Then ω is closed if and only if every point $x \in M$ admits an open neighborhood $\mathcal{U}$ such that $\omega|_{\mathcal{U}} = i\partial\bar{\partial}u$ for some real function u on $\mathcal{U}$.*

Actually, when M is compact one is able to state a global more general result, but we need Hodge theory first.

3.2 Kähler manifolds

Kähler manifolds lie in the intersection between many fields of geometry and their mixed nature makes them definitely worthy of interest. They play a huge role in Hodge Theory and we are going to see some aspects of it.
Let us begin with the concept of **Hermitian metric** on an almost complex manifold (M, J), which is a Riemannian metric g such that $g(JX, JY) = g(X, Y)$ for all $X, Y \in \mathfrak{X}(M)$. The **fundamental form** of g is

$$\omega(X, Y) := \frac{1}{2\pi} g(JX, Y).$$

The extension (denoted g as well) of the Hermitian metric to $TM^{\mathbb{C}}$ by $\mathbb{C}$-linearity satisfies

$$\begin{cases} g(\overline{W}, \overline{Z}) = \overline{g(Z, W)}, \text{ for any } Z, W \in TM^{\mathbb{C}} \\ g(Z, \overline{Z}) > 0, \text{ for any } Z \in TM^{\mathbb{C}} - \{0\} \\ g(Z, \overline{W}) = 0, \text{ for any } Z, W \in T^{1,0}M \text{ and for any } Z, W \in T^{0,1}M. \end{cases}$$

Notice that any complex manifold admits Hermitian metrics: just choose some Riemannian metric $\tilde{g}$ and define $g(X,Y) := \tilde{g}(X,Y) + \tilde{g}(JX,JY)$.

Remark 3.2.1. *Let z^α be homolorphic coordinates on a complex manifold (M^{2m}, J) endowed with an Hermitian metric g and denote $g_{\alpha\bar{\beta}}$ the coefficients of the metric tensor in such coordinates:*

$$g_{\alpha\bar{\beta}} = g\left(\frac{\partial}{\partial z^\alpha}, \frac{\partial}{\partial \bar{z}^\beta}\right).$$

Thus one can check that the fundamental form is given by

$$\omega = \frac{i}{2\pi} \sum_{\alpha,\beta=1}^{m} g_{\alpha\bar{\beta}} dz^\alpha \wedge d\bar{z}^\beta.$$

At this point we can finally introduce the following

Definition 3.2.2. An Hermitian metric g on an almost complex manifold (M, J) is called a **Kähler metric** if J is a complex structure and the fundamental form ω is closed. Thus the manifold M will be called **Kähler manifold**.

In particular one should notice that *Kähler manifolds are symplectic manifolds*, and we aim to describe how, in the Kähler case, some results regarding moment maps allow for an easier formulation. One can take obviously the fundamental form as symplectic form over the manifold: nondegeneracy follows from the hermitian metric being positive definite.
The two following results allow for an expression of the Kähler condition in terms of the covariant derivative of the Levi-Civita connection of g.

Lemma 3.2.3. *Let g be a Hermitian metric on an almost complex manifold (M, J) with Levi-Civita covariant derivative ∇. Then J defines a complex structure on M if and only if*

$$(\nabla_{JX} J)Y = J(\nabla_X J)Y \text{ for all } X, Y \in \mathfrak{X}(M).$$

By mean of this lemma, one can prove that

Theorem 3.2.4. *A Hermitian metric g on an almost complex manifold is Kähler if and only if J is parallel with respect to the Levi-Civita connection of g.*

Recall now that on a Riemannian manifold are defined many useful tensors that tell a great deal about the structure of the manifold. In our situation one can define such tensors using the smooth structure of a Kähler manifold and extend them $\mathbb{C}$-linearly.
Let (M, g, J) be a Kähler manifold. As (M, g) is Riemannian, then we have the following tensors:

- **curvature tensor**:

$$R(X,Y)Z := \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z$$

  for any $X, Y, Z \in \mathfrak{X}(M)$;

- **Riemann tensor**:

$$R(X,Y,Z,W) := g(R(X,Y)Z, W)$$

  for any $X, Y, Z, W \in \mathfrak{X}(M)$;

- **Ricci tensor**:

$$\mathrm{Ric}(X,Y) := \mathrm{Tr}\{V \mapsto R(V,X)Y\},$$

  or equivalently

$$\mathrm{Ric}(X,Y) := \sum_{i=1}^{2m} R(e_i, X, Y, e_i),$$

  for $\{e_i\}_i$ local orthonormal frame of TM.

For the symmetries and properties of these tensors we refer to [23] and [16]. In particular one can check some nice symmetries regarding their composition with the complex structure J.

Definition 3.2.5. The **Ricci form** ρ of a Kähler manifold is defined by

$$\rho(X,Y) := \frac{1}{2\pi} \mathrm{Ric}(JX, Y) \text{ for all } X, Y \in \mathfrak{X}(M).$$

Remark 3.2.6. *The Ricci form plays an important role in the theory to come and let us observe that it is closed. Indeed one exploits the relation*

$$\mathrm{Ric}(X,Y) = \frac{1}{2} \mathrm{Tr}(R(X, JY) \circ J)$$

and computes

$$2d\rho(X,Y,Z) = 2((\nabla_X\rho)(Y,Z) + (\nabla_Y\rho)(Z,X) + (\nabla_Z\rho)(X,Y)) =$$

$$\frac{1}{2\pi}\Big(\operatorname{Tr}((\nabla_X R)(Y,Z)) \circ J + (\nabla_Y R)(Z,X) \circ J + (\nabla_Z R)(X,Y) \circ J)\Big) = 0.$$

These relations show that the only non-zero Christoffel symbols are

$$\Gamma^{\gamma}_{\alpha\beta} \text{ and } \Gamma^{\bar\gamma}_{\bar\alpha\bar\beta}.$$

3.2.1 In local coordinates

One further step in the direction of understanding Kähler manifolds is made by zooming in locally and check how the complex and differential machinery looks like up close. We refer to [23, pp. 33–35].
Let (M, J, g) be a Kähler manifold with the Levi-Civita covariant derivative ∇ and let (z^α) be a system of local holomorphic coordinates. Consider the following local basis of the complexified tangent space:

$$Z_\alpha := \frac{\partial}{\partial z^\alpha}, \quad Z_{\bar\alpha} := \frac{\partial}{\partial z^{\bar\alpha}}, \quad 1 \le \alpha \le m,$$

where $z^{\bar\alpha} := \bar z^\alpha$. Let subscripts $A, B, C, \dots$ run over the set $\{1, \dots, m, \bar 1, \dots, \bar m\}$. We denote the components of the Kähler metric in these coordinates by

$$g_{AB} := g(Z_A, Z_B).$$

Since the metric is hermitian we have

$$g_{\alpha\beta} = g_{\bar\alpha\bar\beta} = 0, \quad g_{\bar\beta\alpha} = g_{\alpha\bar\beta} = \overline{g_{\beta\bar\alpha}}.$$

Let us denote by $g^{\alpha\bar\beta}$ the coefficients of the inverse matrix of $(g_{\alpha\bar\beta})$. The Christoffel symbols are defined by

$$\nabla_{Z_A} Z_B = \Gamma^C_{AB} Z_C.$$

We put $\overline{\overline{\alpha}} = \alpha$ as a convention. By conjugation,

$$\overline{\Gamma^C_{AB}} = \Gamma^{\bar C}_{\bar A\bar B}.$$

Being ∇ torsion-free, we get

$$\Gamma^C_{AB} = \Gamma^C_{BA};$$

since the covariant derivative ∇ along $T^{1,0}$ is zero, we must have

$$\Gamma^\gamma_{A\bar\beta} = 0.$$

The goal now is to compute these coefficients. Notice that $\Gamma^C_{\alpha\bar\delta} = 0$ implies

$$\nabla_{Z_\alpha} Z_{\bar\delta} = 0, \tag{3.1}$$

hence

$$\frac{\partial g_{\beta\bar\delta}}{\partial z^\alpha} = g(\nabla_{Z_\alpha} Z_\beta, Z_{\bar\delta}) = \Gamma^\gamma_{\alpha\beta} g_{\gamma\bar\delta}.$$

This proves the formulas

$$\Gamma^\gamma_{\alpha\beta} g_{\gamma\bar\delta} = \frac{\partial g_{\beta\bar\delta}}{\partial z^\alpha} \text{ and } \Gamma^\gamma_{\alpha\beta} = g^{\gamma\bar\delta}\frac{\partial g_{\beta\bar\delta}}{\partial z^\alpha}. \tag{3.2}$$

Recall that the curvature tensor can be viewed either as $(3,1)$ or as $(4,0)$ tensor. The corresponding coefficients are defined by

$$R(Z_A, Z_B)Z_C = R^D_{ABC} Z_D \text{ and } R_{ABCD} = R(Z_A, Z_B, Z_C, Z_D) = g_{DE} R^E_{ABC}.$$

Being $T^{1,0}M$ ∇-parallel, we get immediately $R^\gamma_{AB\bar\delta} = R^{\bar\gamma}_{AB\delta} = 0$, hence $R_{AB\gamma\delta} = R_{AB\bar\gamma\bar\delta} = 0$. Using the curvature symmetries we finally see that the only non-vanishing components of R are

$$R_{\alpha\bar\beta\gamma\bar\delta}, \quad R_{\alpha\bar\beta\bar\gamma\delta}, \quad R_{\bar\alpha\beta\gamma\bar\delta}, \quad R_{\bar\alpha\beta\bar\gamma\delta}$$

and

$$R^\delta_{\alpha\bar\beta\gamma}, \quad R^{\bar\delta}_{\alpha\bar\beta\bar\gamma}, \quad R^\delta_{\bar\alpha\beta\gamma}, \quad R^{\bar\delta}_{\bar\alpha\beta\bar\gamma}.$$

By applying (3.1) and (3.2) we get

$$R^\delta_{\alpha\bar\beta\gamma} Z_\delta = R(Z_\alpha, Z_{\bar\beta})Z_\gamma = -\nabla_{Z_{\bar\beta}}(\nabla_{Z_\alpha Z_\gamma}) =$$

$$-\nabla_{Z_{\bar\beta}}(\Gamma^\delta_{\alpha\gamma} Z_\delta) = -\frac{\partial \Gamma^\delta_{\alpha\gamma}}{\partial \bar z^\beta} Z_\delta,$$

therefore

$$\boxed{R^\delta_{\alpha\bar\beta\gamma} = -\frac{\partial \Gamma^\delta_{\alpha\gamma}}{\partial \bar z_\beta}.}$$

By mean of this formula, we can finally compute the components of the Ricci tensor:

$$\mathrm{Ric}_{\gamma\bar{\beta}} = \mathrm{Ric}_{\beta\bar{\gamma}} = R^{A}_{A\bar{\beta}\gamma} = R^{\alpha}_{\alpha\bar{\beta}\gamma} = -\frac{\partial \Gamma^{\alpha}_{\alpha\gamma}}{\partial \bar{z}^{\beta}}.$$

Now, let us denote by d the determinant of the matrix $(g_{\alpha\bar{\beta}})$. By mean of some nasty calculations one may show that

$$\Gamma^{\alpha}_{\alpha\gamma} = \Gamma^{\alpha}_{\gamma\alpha} = g^{\alpha\bar{\delta}} \frac{\partial g_{\alpha\bar{\delta}}}{\partial z^{\gamma}} = \frac{1}{d}\frac{\partial d}{\partial z^{\gamma}} = \frac{\partial \log d}{\partial z^{\gamma}}.$$

This proves the following expressions for the Ricci tensor

$$\boxed{\mathrm{Ric}_{\alpha\bar{\beta}} = -\frac{\partial^2 \log d}{\partial z^{\alpha}\partial \bar{z}^{\beta}}}$$

and for the Ricci form

$$\boxed{\rho = -i\partial\bar{\partial}\log d.}$$

3.3 Hodge Theory in short

Hodge theory is mainly concerned with the study of Riemannian and Kähler manifolds from a topological point of view by mean of differential methods: French mathematician Élie Cartan in the first half of the XX century was wondering whether there was any relation between topology and differential forms. It turns out that he was right, and many brilliant mathematicians devoted themselves to the development of this new branch of geometry.
We are mainly interested in the basics of this fascinating theory, so we won't dive too deep into it. As always, one can build up the following machinery on any arbitrary (oriented and euclidean) vector space, but we want to get things going and so we speed up by giving all the definition straight on manifolds.
Our main sources here are [18, pp. 40–49] and [23, pp. 49–52].

Let (M^n, g) be an oriented Riemannian manifold with volume form dv. Let $\{e_1, \ldots, e_n\}$ be a local orthonormal basis, parallel at a point p on M with respect to the Levi-Civita connection ∇. Through the isomorphism between TM and T^*M, we can extend g to 1-forms by setting

$$g(\omega, \eta) = g(\omega^{\sharp}, \eta^{\sharp}).$$

Therefore, if locally $\omega = \sum_{i=1}^m \omega_i e_i^\flat$ and $\eta = \sum_{i=1}^m \eta_i e_i^\flat$, then

$$g(\omega, \eta) = \sum_{i=1}^m \omega_i \eta_i.$$

It is possible to further extend g to k-forms by setting, if $\omega_1, \ldots, \omega_n, \eta_1, \ldots, \eta_n \in \Omega^1 M$,

$$g(\omega_1 \wedge \ldots \wedge \omega_n, \eta_1 \wedge \ldots \wedge \eta_n) = \det\left((g(\omega_i, \eta_j))_{ij}\right).$$

Through these definitions, $\{e_i^\flat \wedge \ldots \wedge e_k^\flat \mid 1 \leq i_1 < \ldots < i_k \leq m\}$ forms a locally orthonormal basis with respect to g for $\Omega^k M$. Therefore, g_x defines a scalar product on $\Omega_x^1 M$ for each $x \in M$. To define a scalar product on $\Omega^k M$, assuming that M is compact, we can set

$$(\omega, \eta) := \int_M g(\omega, \eta)\, dv, \quad \forall \omega, \eta \in \Omega^k M.$$

From the previous definitions, we understand that the exterior and interior products of differential forms are each other's *adjoints*, in the sense that

$$g(\iota_X \omega, \tau) = g(\omega, X^\flat \wedge \tau), \quad \forall X \in \mathfrak{X}(M),\, \omega \in \Omega^k M,\, \tau \in \Omega^{k-1} M.$$

In this setup we define the **Hodge operator** $* : \Omega^k M \to \Omega^{n-k} M$ by mean of the following equation:

$$\omega \wedge *\tau = g(\omega, \tau)\, dv, \quad \forall \omega, \tau \in \Omega^k M,$$

or equivalently,

$$(\omega, \tau) = \int_M \omega \wedge *\tau.$$

We can introduce now the operator $d^* : \Omega^{k+1} M \to \Omega^k M$ as the *formal adjoint* of d, meaning that

$$(d\alpha, \beta) = (\alpha, d^*\beta), \quad \forall \alpha \in \Omega^k M,\, \beta \in \Omega^{k+1} M.$$

One may check that, explicitly,

$$d^* = -(-1)^{nk} * d *.$$

Let us define the **Laplacian operator** $\Delta : \Omega^k M \to \Omega^k M$ as $\Delta := dd^* + d^* d$. The Laplacian operator is *self-adjoint*, meaning that

$$(\Delta\omega, \tau) = (\omega, \Delta\tau), \quad \forall \omega, \tau \in \Omega^k M.$$

The Riemannian framework was a warm-up, since we actually desire to deal with Kähler manifolds. So suppose (M, g, J) is a Kähler manifold.
The extension to k-forms made for a Riemannian metric can be further extended to the complexified exterior bundle, requiring that it defines a Hermitian inner product pointwise. Assuming to denote this extension by $\langle \cdot, \cdot \rangle$ and extending it for $\mathbb{C}$-linearity, then

$$\langle \omega, \tau \rangle = g(\omega, \bar{\tau}).$$

Extended for $\mathbb{C}$-linearity, the Hodge operator satisfies the property

$$\omega \wedge *\bar{\tau} = \langle \omega, \tau \rangle \, dv.$$

In this way, the scalar product of k-forms becomes

$$(\omega, \tau) = \int_M \langle \omega, \tau \rangle \, dv = \int_M \omega \wedge *\bar{\tau}.$$

Remark 3.3.1. *One may check that the Hodge operator $*$ is an operator*

$$* : \Lambda^{p,q} M \to \Lambda^{m-q,m-p} M.$$

Let $\partial^* : \Lambda^{p,q} M \to \Lambda^{p-1,q} M$ and $\bar{\partial}^* : \Lambda^{p,q} M \to \Lambda^{p,q-1} M$ be the formal adjoints of ∂ and $\bar{\partial}$ with respect to $(\cdot, \cdot)$, respectively. Then, $d^* = \partial^* + \bar{\partial}^*$ and,

$$\partial^* = - * \bar{\partial} * \quad \text{and} \quad \bar{\partial}^* = - * \partial * .$$

Indeed, if $\alpha \in \Lambda^{p,q} M$ and $\beta \in \Lambda^{p,q+1} M$,

$$(\bar{\partial}\alpha, \beta) = \int_M \bar{\partial}\alpha \wedge *\beta = \int_M \bar{\partial}(\alpha \wedge *\beta) - (-1)^{p+q} \int_M \alpha \wedge \bar{\partial}^* \beta =$$

$$- \int_M \alpha \wedge *\bar{\partial} * \beta = (\alpha, - * \bar{\partial} * \beta).$$

The case for ∂ is analogous. Therefore, we can define the Laplace operators

$$\boxed{\Delta_\partial := \partial\partial^* + \partial^*\partial} \tag{3.3}$$

and

$$\boxed{\Delta_{\bar{\partial}} := \bar{\partial}\bar{\partial}^* + \bar{\partial}^*\bar{\partial}.} \tag{3.4}$$

So far, one may have noticed that the requirement for M to be complex is not necessary. Indeed the machinery previously defined can be brought up on more general almost Hermitian manifolds endowed with a Kähler form. However, the Kähler case turns out to be lucky enough to allow for the following

Theorem 3.3.2. *Whenever (M, g, J) is a compact Kähler manifold, it holds*

$$\Delta = 2\Delta_\partial = 2\Delta_{\bar\partial}.$$

The reader interested in its proof may find it in [18].
Now we come to the core of Hodge theory. We will discuss cohomology and see some nice properties brought up in this setting.
Assume that (M^n, g) is a compact oriented Riemannian manifold. Let's denote the space of smooth complex-valued k-forms as $\Omega^k M := \Gamma(\Lambda^k M \otimes \mathbb{C})$, and $Z^k M$ as the space of closed complex k-forms on M. Since the exterior derivative satisfies $d^2 = 0$, it is clear that $d\Omega^{k-1}M \subset Z^k M$. We define the **De Rham cohomology groups** as:

$$H^k_{\mathrm{DR}}(M, \mathbb{C}) := \frac{Z^k M}{d\Omega^{k-1}M}.$$

The *De Rham Theorem* states that the k-th cohomology group of M with complex coefficients is naturally isomorphic to the k-th De Rham cohomology group:

$$H^k(M, \mathbb{C}) \cong H^k_{\mathrm{DR}}(M, \mathbb{C}).$$

Now let $H^k(M, \mathbb{C})$ denote the space of complex **harmonic** k-forms on M, i.e. forms in the kernel of the Laplace operator:

$$\mathcal{H}^k(M, \mathbb{C}) := \{\omega \in \Omega^k M | \Delta\omega = 0\}.$$

Lemma 3.3.3. *A form is harmonic if and only if it is closed and d^*-closed.*

Proof. One direction is straightforward. Conversely, let's assume that ω is harmonic. Since M is compact and d and δ are formally adjoint operators, we have

$$0 = \int_M \langle \Delta\omega, \omega\rangle \, dv = \int_M \langle d\delta\omega + \delta d\omega, \omega\rangle \, dv = \int_M |\delta\omega|^2 + |d\omega|^2 \, dv,$$

which shows that $d\omega = 0$ and $\delta\omega = 0$. □

Theorem 3.3.4 (Hodge decomposition Theorem)**.** *The space of k-forms decomposes as a direct sum:*

$$\Omega^k M = \mathcal{H}^k(M, \mathbb{C}) \oplus d^*\Omega^{k+1}M \oplus d\Omega^{k-1}M$$

One of the directions of the proof is a bit hard and goes beyond the interest of this thesis. For the easier direction we refer to [23, p. 49].

The Hodge decomposition Theorem reveals that every k-form ω on M can be uniquely expressed as

$$\omega = d\omega' + d^*\omega'' + \omega^H,$$

where $\omega' \in \Omega^{k-1}M$, $\omega'' \in \Omega^{k+1}M$, and $\omega^H \in \mathcal{H}^k(M, \mathbb{C})$. If ω is closed, we can express

$$0 = (d\omega, \omega'') = (dd^*\omega'', \omega'') = \int_M |d^*\omega''|^2 \, dv,$$

showing that the second term in the Hodge decomposition is zero.
The following is a nice result and allows for an identification of differential forms with the armonic ones:

Theorem 3.3.5 (Hodge isomorphism)**.** *The natural map $f : \mathcal{H}^k(M, \mathbb{C}) \to H^k_{DR}(M, \mathbb{C})$ given by $\omega \mapsto [\omega]$ is an isomorphism.*

Proof. Notice that f is well-defined since every harmonic form is closed. The kernel of f is zero: indeed spaces of harmonic forms and exact forms are orthogonal, so in particular their intersection is $\{0\}$. Finally, for every De Rham cohomology class c, there exists a closed form ω such that $[\omega] = c$. We have seen that the Hodge decomposition of ω is $\omega = d\omega_0 + \omega_H$, showing that $f(\omega_H) = [\omega_H] = [d\omega' + \omega_H] = [\omega] = c$, hence f is surjective.

□

One more fact that we find interesting and moreover useful to this work is the following.

Proposition 3.3.6. *Every Killing vector field on a compact Kähler manifold is real holomorphic.*

Now we mimic the previous construction by mean of $\bar{\partial} : \Omega^{p,q}M \to \Omega^{p,q+1}M$, which is sometimes called the **Dolbeault operator**. Let (M^{2m}, J, g) be an Hermitian manifold.
Let $Z^{p,q}M$ denote the space of $\bar{\partial}$-closed (p,q)-forms. Since $\bar{\partial}^2 = 0$, we see that $\bar{\partial}\Omega^{p,q-1}M \subset Z^{p,q}M$. We define the **Dolbeault cohomology groups**

$$H^{p,q}M := \frac{Z^{p,q}M}{\bar{\partial}\Omega^{p,q-1}M}.$$

In contrast to De Rham cohomology, Dolbeault cohomology is no longer a topological invariant of the manifold, as it depends on J. We define the space $H^{p,q}M$ of **$\bar{\partial}$-harmonic** (p,q)-forms on M by

$$\mathcal{H}^{p,q}M := \{\omega \in \Omega^{p,q}M | \Delta_{\bar{\partial}}\omega = 0\}.$$

Again, we have

Lemma 3.3.7. *A form $\omega \in \Omega^{p,q}M$ is $\bar{\partial}$-harmonic if and only if $\bar{\partial}\omega = 0$ and $\bar{\partial}^*\omega = 0$.*

Similarly as we did before, we present the following result.

Theorem 3.3.8 (Dolbeault decomposition Theorem)**.** *The space of (p,q)–forms decomposes as a direct sum*

$$\Omega^{p,q}M = \mathcal{H}^{p,q}(M,\mathbb{C}) \oplus \bar{\partial}^*\Omega^{p,q+1}M \oplus \bar{\partial}\Omega^{p,q-1}M.$$

But then this implies once more that every (p,q)-form ω on M can be uniquely written as

$$\omega = \bar{\partial}\omega' + \bar{\partial}^*\omega'' + \omega^H,$$

where $\omega' \in \Omega^{p,q-1}M$, $\omega'' \in \Omega^{p,q+1}M$, and $\omega_H \in H^{p,q}M$. This is called the Dolbeault decomposition of ω. The second summand in the Dolbeault decomposition of ω vanishes if and only if $\bar{\partial}\omega = 0$.
In particular, one can show that

Proposition 3.3.9. *A $(p,0)$-form on a compact Hermitian manifold is holomorphic if and only if it is $\bar{\partial}$-harmonic.*

Also, it follows the following usual correspondence:

Corollary 3.3.10 (Dolbeault isomorphism)**.** *The map $f : \mathcal{H}^{p,q}M \to H^{p,q}M$ given by $\omega \mapsto [\omega]$ is an isomorphism.*

We finally conclude this section with some relevant results that are going to be used in Chapter 4, when building up the *Futaki invariant* and the moment map in the case of Kähler Ricci Soliton.

Proposition 3.3.11 (Global $\partial\bar{\partial}$-Lemma)**.** *Let α be a real exact (p,p)-form on a compact Kähler manifold. Then there exists a real $(p-1,p-1)$-form η such that*

$$\alpha = i\partial\bar{\partial}\eta.$$

The proof is a easy check performed by using the previously stated decompositions.

Lemma 3.3.12. *If $X = \operatorname{grad} u$ is a holomorphic vector field, then the real part of X is a Killing vector field if and only if $u+\bar{u}$ is constant. In particular, if u is pure imaginary the real part of* $\operatorname{grad} u$ *is a Killing vector field.*

We refer to [8] for a proof.

3.4 The first Chern class

In differential topology one most powerful tool is represented by *characteristic classes*. These are peculiar cohomology classes that are defined in all generality on fiber bundles (most of the times of vector spaces) and serve to establish important relations. Some examples are Chern classes, the Thom class, the Euler class, Todd classes and others. There are many approaches on how to define them, either by starting with a set of axioms and only afterwards finding who is the candidate class or by giving definitions and proving properties in the orthodox way. In any case, we are not concerned with the tools and results coming from the striking methods of differential topology, so we limit ourselves to the most basic and practical definition of *first* Chern class, which plays some role in the next chapters. Again, we follow [23].

We define the **first Chern Class** of a Kähler manifold (M, J, g) as the class of the real differential form

$$\frac{i}{2\pi}\operatorname{Tr}(\mathrm{Ric}) \in \Omega^2(M, \mathbb{R}),$$

where Ric is as always associated to the Levi-Civita connection ∇. Actually one can prove that the first Chern class is closed, so that

$$c_1(M) := \left[\frac{i}{2\pi}\operatorname{Tr}(\mathrm{Ric})\right] \in H^2_{DR}(M, \mathbb{R}).$$

In particular, notice that $c_1(M)$ is a $(1,1)$-form.

Proposition 3.4.1. *The first Chern class satisfies the following properties:*

1. *it does not depend on the choice of connection on TM;*

2. *whenever M is compact,*

$$c_1(M) = [\rho] \in H^2_{DR}(M, \mathbb{R}),$$

i.e. the Ricci form belongs to the same equivalence class of the first Chern class.

The second point will play an important role in the following. It is also interesting in its own existence. In fact, mathematicians Eugenio Calabi and Shing-Tung Yau have come with a famous result sometimes referred to as the *Calabi conjecture*.

Theorem 3.4.2 (Calabi conjecture)**.** *Let M^{2m} be a compact Kähler manifold with fundamental form φ and Ricci form ρ. Then, for every closed real $(1,1)$-form ρ_1 in the cohomology class $c_1(M)$, there exists a unique Kähler metric with fundamental form φ_1 in the same cohomology class as φ, whose Ricci form is exactly ρ_1.*

A $(1,1)$-form τ on M is said to be **positive** (resp. **negative**) if it is locally expressed by

$$\tau = i\tau_{\alpha\bar{\beta}} dz^\alpha \wedge dz^{\bar{\beta}}$$

with $\tau_{\alpha\bar{\beta}}$ a positive (resp. negative) definite Hermitian matrix: hence τ is necessarily a real form. Similarly, a de Rham class of type $(1,1)$ is said to be positive (resp. negative) if it is represented by a positive (resp. negative) form.
In literature, those Kähler manifolds with positive first Chern class are called **Fano manifolds**. Sometimes we will write $c_1(M) > 0$ as a shorthand for a manifold whose first Chern class is positive.
As a consequence of the Calabi conjecture together with few more machinery, one is able to prove that

Corollary 3.4.3. *Compact Fano manifolds are simply connected.*

Chapter No. 4

KEs and KRSs

The time has come for us to deal with two very special examples of Kähler manifolds. We begin by justifying the interest in Kähler Einstein and Kähler Ricci Soliton metrics and whence this interest comes. We will mainly refer to [30] and [5]. After the structural machinery is outlined, we study two obstructions to the existence of Kähler Einstein and Kähler Ricci Soliton metrics on a given Kähler manifold. We will be adherent to [8] and [29] respectively. Finally, we will be concerned with actions of Lie groups over Kähler Einstein and Kähler Ricci Soliton manifolds. Here we follow [8] in order to write down the moment map for actions over Fano manifolds and we go a bit further: we compute the explicit formula of the moment map when the metric is Kähler Ricci Soliton. With this in hand, we are able to compare it with the *canonical* moment map found, for instance, in [25].

4.1 The basics

Let us introduce the framework where Kähler Einstein and Kähler Ricci Soliton manifolds come from. Consider a Riemannian manifold (M, g) which is compact and without boundary. A powerful tool in differential geometry is the **Ricci flow**, a means of processing the metric g by allowing it to evolve under the following PDE:

$$\boxed{\frac{\partial g}{\partial t} = -2\mathrm{Ric}(g).} \tag{4.1}$$

As already introduced in Chapter 3, we denote by $\mathrm{Ric}(g)$ the Ricci tensor associated to the metric g.

In simple situations, the flow can be used to deform g into a metric distinguished by it curvature. In fact, the topology of M may preclude the existence of such distinguished metrics, and the Ricci flow can then be expected to develop singularities in finite time. Nevertheless, the behaviour of the flow may still serve to tell us much about the topology of the underlying manifold. We present this precious tool since it allows us to introduce the two peculiar metrics we are interested in.
If we take a metric g_0 such that

$$\mathrm{Ric}(g_0) = cg_0$$

for some constant $c \in \mathbb{R}$, then a solution $g(t)$ of (4.1) with $g(0) = g_0$ is given by

$$g(t) = (1 - 2ct)g_0.$$

In particular, for the round unit sphere $(\mathbb{S}^n, g_0)$, we have $\mathrm{Ric}(g_0) = (n-1)g_0$, so that the evolution is $g(t) = (1 - 2(n-1)t)g_0$ and the sphere collapses via the flow to a point at time $T = \frac{1}{2(n-1)}$.
For what is our concern, we move from the pure Riemannian context to the complex one and we start from a Kähler manifold (M, J, g). We define it to be a **Kähler Einstein manifold** (KE, for brevity's sake) if there exists some constant $c \in \mathbb{R}$ such that

$$\boxed{\mathrm{Ric}(g) = cg.}$$

More generally, we define (M, J, g) to be a **Kähler Ricci Soliton** (KRS, for brevity's sake) if there exist some constant $c \in \mathbb{R}$ and some real holomorphic vector field [1] $Y \in \mathfrak{X}(M)$ such that

$$\boxed{\mathrm{Ric}(g) + \frac{1}{2}\mathcal{L}_Y h = cg.}$$

We observe that Kähler Einstein manifolds are a particular case of Kähler Ricci Solitons, for which $Y = 0$. Kähler Ricci Solitons are studied too in differential geometry because of their interesting behaviour under the Ricci flow. Notice that in the definition it is not actually required that M is compact or boundary-less.

[1] Recall Definition 3.1.6.

Kähler Ricci Solitons receive different names depending on the sign of the constant c. If $c = 0$ then the solitons are said to be **steady**, for $c > 0$ they are **shrinking** and when $c < 0$ they are **expanding**. Moreover, if the vector field Y is the gradient of a real valued function f, then the KRS is called itself **gradient** and f has the name of **potential**.

Example 4.1 (The cigar soliton)**.** *Obviously one can give corresponding definitions whenever (M, g) is only a Riemannian manifold. We want to discuss briefly an example. Let $M = \mathbb{R}^2$ and write $g_0 = \rho^2(dx^2 + dy^2)$, for $\rho \neq 0$. The formula for the Gauss curvature is*

$$K = -\frac{1}{\rho^2}\Delta \ln \rho,$$

where here we are writing $\Delta = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$, and the Ricci curvature can be written in terms of the Gauss curvature as $Ric(g_0) = Kg_0$. If now we set

$$\rho^2 = \frac{1}{1 + x^2 + y^2},$$

then we find that $K = \frac{2}{1+x^2+y^2}$, that is,

$$\mathrm{Ric}(g_0) = \frac{2}{1 + x^2 + y^2} g_0.$$

Meanwhile, if we define Y to be the radial vector field $Y := -2(x\frac{\partial}{\partial x} + y\frac{\partial}{\partial y})$, then one can calculate that

$$\mathcal{L}_Y g_0 = -\frac{4}{1 + x^2 + y^2} g_0.$$

Therefore, g_0 flows as a steady ($\lambda = 0$) Ricci soliton. It is illuminating to write g_0 in terms of the geodesic distance s from the origin and of the polar angle θ:

$$g_0 = ds^2 + \tanh^2 s \, d\theta^2.$$

This shows that the cigar opens at infinity like a cylinder and therefore looks like a cigar. It is useful to know the expression of the Gauss curvature in these coordinates:

$$K = \frac{2}{\cosh^2 s}.$$

Finally, note that the cigar is also a gradient soliton since Y is radial. Indeed, we may take $f = -2 \ln \cosh s$.

Most of the time we will work in coordinates, thus let us express the KE and KRS equations with the same notations of Chapter 3:

$$R_{\alpha\bar{\beta}} = cg_{\alpha\bar{\beta}} \tag{4.2}$$

$$R_{\alpha\bar{\beta}} + \frac{1}{2}(\nabla_{\bar{\beta}}Y_\alpha + \nabla_\alpha Y_{\bar{\beta}}) = cg_{\alpha\bar{\beta}} \tag{4.3}$$

We will be mostly interested in the gradient case, where $Y = \nabla f$. In such case one writes (4.3) down as

$$R_{\alpha\bar{\beta}} + \nabla_\alpha\nabla_{\bar{\beta}}f = cg_{\alpha\bar{\beta}} \quad \text{and} \quad \nabla_\alpha\nabla_\beta f = 0. \tag{4.4}$$

For what concerns the goal of this work, from now on assume M to be compact.

Lemma 4.1.1 (Hamilton)**.** *Let g be a complete gradient Ricci Soliton with potential function f. Then we have*

$$S + ||\nabla f||^2 - 2cf = K \tag{4.5}$$

for some constant K. Here S denotes the scalar curvature.

We refer to [5, p.2] for its quick proof. Suppose also M to be a KRS and consider the following

Proposition 4.1.2 (cf. [5])**.** *Any compact steady or expanding KRS must be Einstein.*

Proof. Taking the trace in equation (4.4), we get

$$S + \Delta f = nc. \tag{4.6}$$

Taking the difference of (4.5) in Lemma 4.1.1 and (4.6), we get

$$\Delta f - ||\nabla f||^2 + 2cf = nc - K.$$

When M is compact and $c \leq 0$, it follows from the maximum principle that f must be a constant, and hence M is KE. □

Then we can take the constant c in (4.3) to be positive, i.e. we deal with a shrinking KRS. Actually, we can even say more. Indeed Grigori Perelman showed, as side effect of more powerful results, that any compact shrinking Ricci soliton is necessarily a gradient soliton (cf. [5, p.7]). Thus so far we have achieved that a compact KRS (M, J, g) is also

- *gradient*: $Y = \nabla f$
- *shrinking*: $c > 0$

Let us simplify even further and notice that we can normalize the constant, i.e. take $c = 1$. Indeed the Christoffel symbols are invariant by scaling and so is the Ricci tensor, whose local expression is

$$R_{\alpha\bar{\beta}} = -\frac{\partial^2 \det(g_{\gamma\bar{\delta}})}{\partial z^\alpha \partial z^{\bar{\beta}}}.$$

Then we rewrite the normalized KRS equation as

$$R_{\alpha\bar{\beta}} + \nabla_\alpha \nabla_{\bar{\beta}} f = g_{\alpha\bar{\beta}}. \tag{4.7}$$

It is noticeable that one can equivalently write this structure equation by mean of the Ricci form ρ instead of Ric and the fundamental form ω instead of g. Moreover, since

$$\mathcal{L}_{\nabla f} g = \partial\bar{\partial} f,$$

in this peculiar case equation (4.3) becomes

$$\rho + \frac{i}{2\pi}\partial\bar{\partial} f = \omega. \tag{4.8}$$

4.2 The Futaki invariant

An interesting problem in the field of complex differential geometry is to determine whether a Kähler manifold (M, J, g) admits an Einstein metric or not. Very often the first Chern class of the manifold can tell us a great deal on this matter.
From now on assume M to be compact.
Different solutions and obstructions had been given depending on the first Chern class of M being positive or negative.

Proposition 4.2.1. *Let M be a compact Kähler manifold with its Kähler form ω. If $\rho' = \frac{i}{2\pi} R'_{ij} dz^i \wedge \overline{dz^j}$ is a real closed $(1,1)$-form which represents the first Chern class $c_1(M)$, then there exists a unique Kähler form ω' in the cohomology class $[\omega]$ such that the Ricci form of ω' is equal to ρ'.*

Now, suppose that $c_1(M) > 0$. Proposition 4.2.1 together with the second point of Proposition 3.4.1 implies that both ρ and ω belong to the equivalence class $c_1(M)$. Hence, the global $\partial\bar{\partial}$-Lemma 3.3.11 gives that there exists a smooth function F such that

$$\boxed{\rho - \omega = \frac{i}{2\pi}\partial\bar{\partial}F.} \tag{4.9}$$

Let $\mathfrak{h}(M)$ be the Lie algebra which consists of all holomorphic vector fields on M. Suppose that $\dim_{\mathbb{C}}(M) = n$. We define a linear function $\eta : \mathfrak{h}(M) \to \mathbb{C}$ by

$$\eta(X) := \frac{1}{2\pi}\int_M X(F)\omega^n.$$

This is called the **Futaki invariant**.

Theorem 4.2.2. *The definition of η does not depend on the choice of the Kähler form and the metric. In particular, η is invariant under the coadjoint action of the group of all automorphisms of M; in particular again, η is a Lie algebra homomorphism. If M admits a KE metric, then η vanishes identically.*

4.3 The Tian invariant

For the sake of completeness, in this section we describe briefly an obstruction to the existence of KRS metrics. In the following we will be adherent to [29]. As always, let (M, J, g) be a compact Kähler manifold and furthermore let us assume that $c_1(M) > 0$. Thanks to the $\partial\bar{\partial}$-Lemma, we have

$$\rho - \omega = \frac{i}{2\pi}\partial\bar{\partial}F \tag{4.10}$$

for some smooth function F.
Due to Corollary 3.4.3 M is simply connected and hence for any holomorphic vector field X on M, there is a unique smooth complex-valued function $\vartheta_X(g)$ of M such that

$$\begin{cases} \iota_X\omega = \frac{i}{2\pi}\bar{\partial}\vartheta_X(g) \\ \int_M e^{\vartheta_X(g)}\omega^n = \int_M \omega^n \end{cases} \tag{4.11}$$

We define the linear functional $F_X : \mathfrak{h}(M) \to \mathbb{C}$ by

$$F_X(v) := \int_M v(F - \vartheta_X(g))e^{\vartheta_X(g)}\omega^n.$$

This functional was introduced by Gang Tian in [29] and it is sometimes referred to as **Tian's invariant**. One can show that it is a holomorphic invariant on M and, most importantly, that it represents an obstruction to the existence of KRS metrics over M. It is remarkable that, if M is KE, then by taking $X = 0$ we get the Futaki invariant.
The proof that this new invariant is in fact an invariant is rather complicated and technical and it goes beyond the scope of this thesis; let us prove instead the following more significant theorem:

Theorem 4.3.1. *If M admits a Kähler–Ricci soliton ω with respect to a holomorphic vector field X then the corresponding holomorphic invariant F_X identically vanishes, i.e.*

$$F_X(v) = 0 \textit{ for all } v \in \mathfrak{h}(M).$$

Proof. By invariance of F_X, it suffices to fix a KRS metric h. By comparing equations (4.10) and (4.11) we understand that, since

$$\mathcal{L}_X\omega = \partial\iota_X\omega,$$

it follows

$$\mathcal{L}_X\omega = \frac{i}{2\pi}\partial\bar{\partial}\vartheta_X(g).$$

The maximal principle together with the equation of KRS implies that

$$F - \vartheta_X(g) = const.$$

Now the statement follows directly. □

4.4 Actions over KE and KRS

The purpose of this section is to say something about moment maps when a certain kind of action is performed on manifolds endowed with a KE or KRS metric. By mean of the theory in [8], we give an explicit formula for moment maps on KRS and thus generalize the *canonical* moment map studied for KE in [25].
Let M be a compact Kähler manifold of positive first Chern class, and g be a Kähler metric whose equivalence class coincides with $c_1(M)$. Let ω and ρ be its Kähler and Ricci forms. We already observed the existence of

some smooth function F such that (4.10) holds. We define a second order differential operator Δ_F on $C^\infty(M,\mathbb{C})$ by

$$\Delta_F u = \Delta_{\bar\partial} u - \nabla u(F),$$

where $\Delta_{\bar\partial} = \bar\partial^* \bar\partial$ is the (complex) Laplacian and $u \in C^\infty(M,\mathbb{C})$, as introduced in (3.4). As a shorthand, we set $\Delta := \Delta_{\bar\partial}$. Δ_F is self-adjoint with respect to the inner product defined by

$$(u,w) = \int_M u\bar{w}e^F dv.$$

We say that $\lambda \in \mathbb{C}$ is an **eigenvalue** of Δ_F if there exists a function $u \in C^\infty(M,\mathbb{C})$ not identically zero, such that $\Delta_F u = \lambda u$. Since

$$\int_M \Delta_F u \cdot \bar{u} e^F dv = \int_M |\partial u|^2 e^F dv$$

for any $u \in C^\infty(M,\mathbb{C})$, all the eingenvalues of Δ_F are nonnegative real numbers. Due to the theory of elliptic operators (check [26, p. 517] for more precise analytic statements) is the following property: there is a sequence of eigenvalues $0 = \lambda_0 < \lambda_1 < \dots$, tending to ∞. If Λ_i is the space of eingenfunctions with respect to λ_i, again since Δ_F is elliptic we have that Λ_i is finite dimensional for each i. We recall he following theorem that will be useful in the sequel.

Theorem 4.4.1. *With the notation introduced above, under the same hypothesis we have that $\lambda_1 \geq 1$. If $\lambda_1 = 1$, then Λ_1 is isomorphic to the complex Lie algebra $\mathfrak{h}(M)$ of all holomorphic vector fields on M through the correspondence $u \mapsto \nabla u$.*

Check [8, pp. 41–42] for a proof.
Notice that if g is KE, then F is constant and thus $\Delta_F = \Delta$.
Clearly if $\lambda_1 > 1$ we can normalize the sequence $\{\lambda_j\}$ by setting $\tilde\lambda_j = \lambda_j/\lambda_1$, so that $\tilde\lambda_1 = 1$. Thus in our situation we can always assume that Theorem 4.4.1 holds.
We will say that an action $\psi : G \times M \to M$ of a Lie group G over some Kähler manifold M is **holomorphic** whenever the map $\psi^g : M \to M$ is holomorphic for each $g \in G$. Suppose now that a compact (path) connected

Lie group G acts on (M, J, g) holomorphically and isometrically. Then there is an embedding of Lie algebras $\mathfrak{g} \to \mathfrak{h}(M)$ defined by

$$X \to X' := \frac{1}{2}(\widehat{X} - iJ\widehat{X}),$$

where $\widehat{X}$ is the action field of X, defined in (1.2). By Theorem 4.4.1, there is $u_{X'} \in \Lambda_1$ such that $X' = \nabla u_{X'}$.

Remark 4.4.2. *Notice that $u_{X'} \in \Lambda_1$ is pure imaginary. Let us see why. Since G acts on M by isometries, F is G-invariant; thus $\widehat{X}(F) = 0$. Therefore we have*

$$\Delta_F u_{X'} = \Delta u_{X'} - X'(F) = \Delta u_{X'} + \frac{i}{2}(J\widehat{X})(F)$$

and

$$\overline{\Delta_F u_{X'}} = \overline{\Delta u_{X'}} - \frac{i}{2}(J\widehat{X})(F).$$

Since $\Delta_F u_{X'} = u_{X'}$, we have

$$\Delta(u_{X'} + \overline{u_{X'}}) = u_{X'} + \overline{u_{X'}}.$$

Now the conclusion follows from Lemma 3.3.12. Indeed $Re(X') = \frac{1}{2}\widehat{X}$ which is naturally a Killing vector field. Hence

$$0 = \Delta(u_{X'} + \overline{u_{X'}}) = u_{X'} + \overline{u_{X'}}.$$

We can introduce a moment map for a holomorphic isometric action of a compact (path) connected Lie group G over a compact connected Kähler manifold (M, g, ω, J).
We put

$$\boxed{\mu^X = \frac{i}{2\pi} u_{X'}} \tag{4.12}$$

and define

$$\mu : M \to \mathfrak{g}^*, \quad \langle \mu(p), X \rangle := \mu^X(p)$$

for some fixed scalar product on $\mathfrak{g}$ which is invariant for the adjoint action Ad_g. Notice that, as usual, here we assume implicitly the natural isomorphism $\mathfrak{g}^* \simeq \mathfrak{g}$, so that $\mathfrak{g}^*$ is analogously endowed with an Ad_g^*-invariant scalar product.
Now we show that

Lemma 4.4.3. *$\mu : M \to \mathfrak{g}^*$ is a moment map for the action of G.*

Proof. Since $\omega = \frac{i}{2\pi} g_{\alpha\bar{\beta}} dz^\alpha \wedge d\overline{z^\beta}$, we have

$$\iota_{X'}\omega = \iota_{\nabla u_{X'}}\omega = \frac{i}{2\pi} h^{\alpha\bar{\beta}} \nabla_{\bar{\beta}} u_{X'} h_{\alpha\bar{\gamma}} d\overline{z^\gamma} = \bar{\partial}\mu^X$$

and

$$\iota_{\widehat{X}}\omega = \iota_{X'+\overline{X'}}\omega = \iota_{X'}\omega + \overline{\iota_{X'}\omega} = d\mu^X.$$

Thus we understand that μ^X is a Hamiltonian function for $\widehat{X}$.

Now, choose $h \in G$. Recall that, by mean of the action $\psi : G \times M \to M$, we can define a map $\psi^h : x \mapsto \psi(h, x)$. Thus $(\psi^h)^*$ commutes with Δ_F since it commutes with Δ and $(\psi^h)^* F = F$. Therefore, if $u_{X'} \in \Lambda_1$ so does $(\psi^h)^* u_{X'}$. Hence we can work *by duality* with gradients. For notations' sake define $h^* := (\psi^h)^*$ and $h_* := d\psi^h$; take a vector field Y of type $(0, 1)$. Then,

$$\omega(\nabla(h^* u_{X'}), Y) = \frac{1}{2\pi} g(J\nabla(h^* u_{X'}), Y) = \frac{-1}{2\pi} g(\nabla(h^* u_{X'}), JY)$$
$$= \frac{-1}{2\pi} JY(h^* u_{X'}) = \frac{-1}{2\pi} h_*(JY)(u_{X'}) = \frac{-1}{2\pi} J(h_* Y)(u_{X'}) =: (*),$$

where we used the relation between the symplectic form and the metric, their symmetries and the definition of gradient. Also, in the second row we used the fact that the differential of a holomorphic map commutes with the almost complex structure J, as already observed in Chapter 2.

$$(*) = \frac{-1}{2\pi} g(\nabla u_{X'}, Jh_* Y) = \frac{1}{2\pi} g(J\nabla u_{X'}, h_* Y) = \omega(\nabla u_{X'}, h_* Y)$$
$$= \omega(X', h_* Y) = \omega(h_*^{-1} X', Y) = \omega((\mathrm{Ad}_{h^{-1}} X)', Y)$$
$$= \omega(\nabla u_{(\mathrm{Ad}_{h^{-1}} X)'}, Y).$$

Notice that we used the following facts: X' is the gradient of $u_{X'}$ and the action is by symplectomorphisms. Also, we used the fact that the gradient of $u_{(\mathrm{Ad}_{h^{-1}} X)'}$ is $(\mathrm{Ad}_{h^{-1}} X)'$.

This implies

$$h^* u_{X'} = u_{(\mathrm{Ad}_{h^{-1}} X)'},$$

and thus

$$h^* \mu^X = \mu_{\mathrm{Ad}_{h^{-1}} X}.$$

This condition amounts to saying that

$$\mu \circ \psi^h = \mathrm{Ad}_h^* \circ \mu,$$

so that we proved G-equivariance for μ.
It follows that μ is a moment map for the G-action. □

The previous moment map was introduced and studied by Akito Futaki in [8] for the case $c_1(M) > 0$. More specifically, one may wonder what is its shape when the manifold is KRS. We have found this expression by combining the results established by Futaki with the equation of KRS.
So now, suppose that (M, J, g) is a KRS. By looking at both equations (4.8) and (4.9) we understand that

$$F = -f.$$

Hence

$$\mu^X = \frac{i}{2\pi} u_{X'} = \frac{i}{2\pi} \Delta_F u_{X'} = \frac{i}{2\pi} \Big(\Delta u_{X'} + \nabla u_{X'}(f) \Big).$$

Recall that $\nabla u_{X'} = X' = \frac{1}{2}(\widehat{X} - iJ\widehat{X})$ and that $\Delta\varphi = -\operatorname{div}(\nabla\varphi)$ for any $\varphi \in C^2(M, \mathbb{C})$. Thus we have

$$\mu^X = \frac{i}{2\pi} \left(-\operatorname{div}\left(\frac{1}{2}(\widehat{X} - iJ\widehat{X}) \right) + (\widehat{X} - iJ\widehat{X}) f \right).$$

One can prove that action fields are Killing vector fields, i.e. $\mathcal{L}_{\widehat{X}} g = 0$ for any $X \in \mathfrak{g}$. One only needs to show that their flow is an isometry and this is achieved by recalling that the flow associated to $\widehat{X}$ is $\psi^{\exp(tX)}$ and that the action is by symplectomorphisms. One then exploits the relation between the metric and its associated Kähler form to get to the claim.
Moreover, we have that $\operatorname{div}(\widehat{X}) = 0$ for any $X \in \mathfrak{g}$. Indeed, one applies the definitions of divergence and Lie derivative to get

$$\operatorname{div}(X)\omega = \mathcal{L}_X \omega = \frac{d}{dt}\bigg|_{t=0} (\psi^{\exp(tX)})^* \omega = 0.$$

Also, since f is G-invariant and $\widehat{X}$ is tangent to orbits, we get $\widehat{X}(f) = 0$. Thus, we are left with

$$\mu^X = \frac{i}{2\pi}\left(-\operatorname{div}\left(\frac{1}{2}(-iJ\widehat{X})\right) - (iJ\widehat{X})f\right).$$

We conclude that

$$\boxed{\mu^X = \left(\frac{-\operatorname{div} + df}{4\pi}\right)(J\widehat{X}).} \tag{4.13}$$

In particular, when (M, J, g) is KE, the map F in equation (4.9) can be taken as a constant; thus $\Delta_F = \Delta$. The moment map $\mu : M \to \mathfrak{g}^*$ along some $X \in \mathfrak{g}$ has thus the following expression:

$$\boxed{\mu^X = \frac{-1}{4\pi}\operatorname{div}(J\widehat{X}).} \tag{4.14}$$

Let us check that the moment map works well in a particular case. Consider the $\mathbb{R}$-action generated by the flow of the vector field $X = J\nabla f$, which is Killing[2]. The moment map for this action is particularly simple:

$$\iota_{J\nabla f}\omega = \omega(J\nabla f, \cdot) = -\frac{1}{2\pi}g(\nabla f, \cdot) = -\frac{1}{2\pi}df.$$

So, we expect the moment map defined in (4.13) to coincide with $-\frac{1}{2\pi}f$. A quick check gives

$$\mu^{J\nabla f} = \frac{\Delta f - ||\nabla f||^2}{4\pi}. \tag{4.15}$$

If we now subtract (4.5) to (4.6) and we substitute into (4.15), we obtain

$$\mu^{J\nabla f} = \frac{n - 2f - K}{4\pi} = -\frac{f}{2\pi} + \tilde{K}. \tag{4.16}$$

Since moment maps are determined up to constants, we proved that the moment map along $J\nabla f$ is a multiple of f.

The moment map μ studied by Futaki in [8] is a generalization to Fano manifolds (i.e. those such that $c_1 > 0$) of the *canonical moment map* φ

[2]This is an easy consequence of $\mathcal{L}_{J\nabla f}\omega$ and $\mathcal{L}_{\nabla f}J$ being zero. Notice that the first Lie derivative coincides with $-d^2 f = 0$, while the second vanishes due to ∇f being real holomorphic.

found, for instance, in [25] of Fabio Podestà. There is a slight ambiguity in term of constants, depending on the chosen normalization of the fundamental form of the Kähler manifold, but this uncertainty is a common price to pay in differential geometry. Some authors choose not to normalize the fundamental Kähler form and the Ricci form by a factor of 2π, so whenever checking the references one should be aware of this inconsistency.
Let us focus on [25] and see that we find again the expression in (4.14).

Lemma 4.4.4. *Let $\tilde{\omega}$ be a G-invariant Kähler form. The map $\varphi : M \to \mathfrak{g}^*$ given by*

$$\langle \varphi(p), X \rangle := -\frac{1}{4\pi} \operatorname{div}(J\widehat{X})$$

is G-equivariant and satisfies for $Y \in T_pM$

$$\langle d\varphi_p(Y), Z \rangle = \tilde{\rho}(Y, Z),$$

where $\tilde{\rho}$ is the Ricci form relative to $\tilde{\omega}$. In particular, if M is KE with $\tilde{\rho} = c\tilde{\omega}$, $c \neq 0$, then $\frac{1}{c}\varphi$ is the moment map relative to $\tilde{\omega}$.

Proof. We won't check equivariance, so let us focus on the other condition. Recall that for every vector field X, by definition $\mathcal{L}_X\omega = \operatorname{div}(X)\omega$; if ∇ is the Levi-Civita connection of the corresponding Kähler metric g, then $\operatorname{div}(X)$ is the trace of the tensor ∇X. So, in our situation at a fixed point $p \in M$, if $\{e_1, \dots, e_n\}$ is a local orthonormal frame around p with $\nabla_{e_j} e_k|_p = 0$ for all $j, k = 1, \dots, n$, we have at p:

$$Y(\operatorname{div}(J\widehat{X}_i)) = g(\nabla_Y \nabla_{e_i} J\widehat{X}, e_i) = -g(\nabla_Y(\nabla X)(e_i), Je_i)$$

$$= g(R(\widehat{X}, Y)e_i, Je_i) = -\operatorname{Tr}(R(\widehat{X}, Y) \circ J) = 4\pi\rho(\widehat{X}, Y).$$

□

Chapter No. 5

Homogeneous Lagrangian submanifolds

In this chapter we desire to treat the theory of Lagrangian orbits, mainly following [9]. We begin with some nice advancements made almost fifty years ago by the mathematician Francis C. Kirwan in the study of properties of squared norm of the moment map, which behaves as a sort of Morse-Bott non-degenerate function. This icebreaker allows us to characterize in an interesting way the topology of the Lagrangian orbit sent to zero via the moment map, whenever an open Stein orbit exists for the action. Finally, the third paragraph is devoted to the investigation of the minimality of Lagrangian orbits of KE and KRS. In particular, we analyze two results achieved in [9] in the KE case and moreover characterize in the general case the (unique) Lagrangian orbit as a minimal submanifold.

5.1 The squared norm of the moment map

In this section we briefly go through the material of [14] and understand in which way some results may help us to study homogeneous Lagrangian submanifolds.
Suppose that a Lie group K acts in an Hamiltonian fashion over some symplectic manifold (M, ω) for some moment map $\mu : M \to \mathfrak{k}^*$. We fix an inner product on the Lie algebra $\mathfrak{k}$ which is invariant under the adjoint action of K and denote it by $\langle \cdot, \cdot \rangle$. As usual, we use it to identify $\mathfrak{k}$ with $\mathfrak{k}^*$. Also, suppose that M is endowed with a K-invariant metric g, which is the same as saying that (M, ω, g) is also a Riemannian manifold and the action is isometric.

Definition 5.1.1. Let $f : M \to \mathbb{R}$ be the function given by

$$f(x) = ||\mu(x)||^2$$

for $x \in M$, where $|| \cdot ||$ is the norm on $\mathfrak{k}$ induced by the fixed inner product.

In [14], one wishes to treat this function as a Morse function. A **Morse function** is a smooth function $f : M \to \mathbb{R}$ such that any of its critical points is *non-degenerate*, i.e. if $d_p f = 0$ then the *Hessian* $H_p f : T_p M \to T_p^* M$ is an isomorphism. Recall that, if for each $v \in T_p M$ we define $\tilde{v}$ its extension to a vector field on M, then the Hessian is defined as

$$H_p f(v)(w) := \tilde{v}_p(\tilde{w}(f)) \text{ for any } v, w \in T_p M;$$

notice that $\tilde{v}_p$ is just v. For a gentle introduction to Morse theory we refer for instance to [22]. Whenever we talk about Morse function, one can keep in mind the previous definition. However, one should be aware that [14] uses a slightly different definition, that of Morse function *in the sense of Bott* (also referred to as **Morse-Bott function**, or *non-degenerate Morse function in the sense of Bott*). We refer to [3, p. 341] for further details on its properties. For any $x \in M$, let $\{x_t : t \geq 0\}$ be the trajectory of $-\nabla f$ such that $x_0 = x$, i.e., the path of steepest descent of f starting from x. Define $\nu(x) = \{y \in M :$ every neighborhood of $y \in X$ contains points x_t for t arbitrarily large$\}$, i.e. the set of *limit points* of the trajectory as $t \to \infty$. Then $\nu(x)$ is closed and nonempty (since M is compact) and is connected. Also, each element of $\nu(x)$ is critical for f.
The following construction will sound familiar to the reader who knows about Morse theory, but since our purpose does not include this branch of geometry, we will usually avoid going into details.
We give here the definition of stratification as prescribed in [14], since it differs slightly from the one in [1] that was given in Definition 1.3.8. For some aspects, the following definition is more general and was needed by the author of [14] for dealing with cohomology groups. We remark that our interest is limited to the following definition.

Definition 5.1.2. A finite collection

$$\{S_\beta : \beta \in B\}$$

of subsets forms a **stratification** of M if M is the disjoint union of the strata S_β, and there is a strict partial order $>$ on the indexing set B such that

$$\overline{S}_\beta \subseteq \bigcup_{\gamma \geq \beta} S_\gamma$$

for every $\beta \in B$. The stratification is **smooth** if every S_β is a locally-closed submanifold of X (possibly disconnected).

Sometimes the partial order can be given, for instance, by $C > C'$ if $g(C) > g(C')$, where $g : \mathcal{C} \to \mathbb{R}$, $C, C' \in \mathcal{C}$.
One of the reasons why stratifications are worth of interest is that given any smooth stratification $\{S_\beta : \beta \in B\}$ of the manifold M, one can build up the cohomology of M inductively from the cohomology of the strata. Anyway, let us get back to our situation and assume both K and M to be compact. To begin with, we shall investigate first the critic point for f. Recall that for $\xi \in \mathfrak{g}$, we defined $\widehat{\xi}_x$ to be its associated action field valued in $x \in M$.

Lemma 5.1.3. *A point $x \in M$ is critical for f if and only if $\widehat{\mu(x)}_x = 0$, where $\mu(x) \in \mathfrak{k}$ is identified with an element of $\mathfrak{k}$ by using the fixed invariant inner product on $\mathfrak{k}$.*

Now, recall Theorem 1.2.6; let T be a maximal torus of K, and let $\mathfrak{t}$ be its Lie algebra. It is straightforward to verify that the composition $\mu_T : M \to \mathfrak{k}^* \to \mathfrak{t}^*$ of μ with the restriction map $\mathfrak{k}^* \to \mathfrak{t}^*$ is a moment map for the action of T on M. When the inner product of $\mathfrak{k}$ is used to identify $\mathfrak{k}^*$ with $\mathfrak{k}$ and $\mathfrak{t}^*$ with $\mathfrak{t}$, then μ_T becomes the orthogonal projection of μ onto $\mathfrak{t}$. Thus, if $\mu(x) \in \mathfrak{t}$, then $\mu_T(x) = \mu(x)$, and hence x is critical for the function $f = \|\mu\|^2$ if and only if it is critical for the function $f_T = \|\mu_T\|^2$ by Lemma 5.1.3.
Therefore, we shall next investigate the critical points of f_T. Actually, moment maps $\mu_T : M \to \mathfrak{t}^*$ associated to torus actions have been deeply studied by the mathematician Francis Atiyah, who proved the following delightful result:

Theorem. *The image under μ_T of the fixed point set of T on M is a finite set $\mathcal{A}$ of points in $\mathfrak{t}^*$, and $\mu_T(x)$ is the convex hull $Conv(\mathcal{A})$ of $\mathcal{A}$ in $\mathfrak{t}^*$. $\mathcal{A}$ is called the set of **weights**.*

So really, whenever a torus is acting, the image of the moment map will not be too much complicated: in fact it consists of all the convex combinations

of images of the (finitely many!) fixed points of the action. One may check for instance [27] for some nice drawings.
For any $\beta \in \mathfrak{t}$, let T_β be the closure in T of the real one-parameter subgroup $\exp(\mathbb{R}\beta)$. Thus, T_β is a subtorus of T. Let $\mu_\beta : M \to \mathbb{R}$ be given by $\mu_\beta(x) = \langle \mu(x), \beta \rangle$. Then, by the definition of a moment map, the cotangent field $x \to d\mu_\beta(x)$ on M is ω-dual to the vector field $x \to \widehat{\beta}_x$ induced by β on M. If $x \in M$, then $\widehat{\beta}_x = 0$ if and only if x is fixed by the subgroup $\exp(\mathbb{R}\beta)$ of T and hence by its closure T_β in T. Therefore the critical set of the function μ_β on M is precisely the fixed point set of the subtorus T_β of T.
In particular, Atiyah was able to show that

Proposition 5.1.4. *μ_β is a non-degenerate Morse function on M in the sense of Bott.*

Definition 5.1.5. Let Z_β be the union of those connected components of the critical set of μ_β on which μ_β takes the value $\|\beta\|^2$.

Thus, if $x \in Z_\beta$, then $\mu(x)$ lies in the affine hyperplane in $\mathfrak{k}$ containing β and perpendicular to the line from β to the origin. Z_β is a submanifold of X (possibly disconnected), fixed by T_β, and invariant under T. In fact, it is a symplectic submanifold of X.
These definitions aim to the following results.

Lemma 5.1.6. *Let $x \in M$ and let $\beta = \mu_T(x) \in \mathfrak{t}$. Then, x is critical for $f_T = \|\mu_T\|^2$ if and only if $x \in Z_\beta$; and if this is the case, then β is the closest point to 0 of the convex hull of some nonempty subset of the set of weights $\mathcal{A}$.*

The idea now is to use this lemma so that we can describe the critical set of the function $f = \|\mu\|^2$ associated with the action of the whole group K.

Definition 5.1.7. Let $\beta \in \mathfrak{t}$, the Lie algebra of a maximal torus T of K. Then, β will be called the **minimal combination of weights** of the action of T on X if it is the closest point to the origin of the convex hull in $\mathfrak{t}$ of some nonempty subset of the set of weights $\mathcal{A}$. Let $\mathfrak{t}_+$ be a fixed positive Weyl chamber[3] in $\mathfrak{t}$, and denote by $\mathcal{B}$ the set of all minimal combinations of

[3]For the detailed description on what a (positive) Weyl chamber is, we refer to [1, pp. 93–97]. It is quite a technical notion and we do not mean to overcomplicate the prerequisite section with further definitions. One can consider it as a "slice" of the Lie algebra.

weights which lie in $\mathfrak{t}_+$. $\mathcal{B}$ will be the indexing set for the stratification of M which we shall associate with the function f.

For each $\beta \in \mathcal{B}$, let us define

$$\boxed{C_\beta = K(Z_\beta \cap \mu^{-1}(\beta)).}$$

Then we have

Lemma 5.1.8. *The critical set of f on M is the disjoint union of the closed subsets $\{C_\beta : \beta \in \mathcal{B}\}$ of M.*

We choose not to include any of these proofs due to their technicality and lack of relevance with the main topic of the present thesis. Anyway, the interested reader is referred to [14].
The subsets $\{C_\beta : \beta \in \mathcal{B}\}$ will be called the *critical subsets* for f.

Corollary 5.1.9. *The image under μ of each connected component of the critical set for f is a single adjoint orbit in $\mathfrak{k} \simeq \mathfrak{k}^*$. For each $\beta \in \mathcal{B}$, C_β consists of those critical points for f whose image under μ lies in the adjoint orbit of β. Thus, the function $f = \|\mu\|^2$ takes the value $\|\beta\|^2$ on C_β.*

Kirwan finally shows in her [14] how one is able to find a nice stratification of M. Everything is summarized in the following

Theorem 5.1.10. *Let M be a compact symplectic manifold acted on by a compact Lie group K, and suppose $\mu : X \to \mathfrak{k}$ is a moment map for this action. Fix an invariant inner product on $\mathfrak{k}$. Then the set of critical points for the function $f = \|\mu\|^2$ is a finite disjoint union of closed subsets $\{C_\beta : \beta \in B\}$ on each of which f takes a constant value. There is a smooth stratification*

$$\{S_\beta : \beta \in B\}$$

of M such that a point x in M lies in the stratum S_β if and only if the limit set of the path of steepest descent for f from x (with respect to a suitable K-invariant metric) is contained in C_β.

Even though we didn't check with hands if is indeed a smooth stratification, we want to keep in mind its shape and wish to express it differently in some peculiar cases.

Suppose now that M is a compact Kähler manifold acted on by a complex Lie group G, and that G is the complexification of a maximal compact subgroup K. Thus, if $\mathfrak{k}$ and $\mathfrak{g}$ are the Lie algebras of K and G respectively, then $\mathfrak{g} = \mathfrak{k} \oplus i\mathfrak{k}$. Suppose also that K preserves the Kähler structure on M. This condition is always satisfied if the Kähler metric is replaced by its average over K.
The Kähler structure makes M into a symplectic manifold acted on by K, and in addition, gives M a K-invariant Riemannian metric. Assume that a moment map $\mu : M \to \mathfrak{k}$ exists for the action of K on M.
We define $Z_\beta^{\min} \subset Z_\beta$ as consisting of those points $x \in Z_\beta$ such that the limit points of the path of steepest descent from x for $\|\mu - \beta\|^2$ on Z_β lie in $Z_\beta \cap \mu^{-1}(\beta)$.
Analogously, define Y_β as the subset of M consisting of all those points in M whose paths of steepest descent under μ_β converge to points of Z_β; one can prove (see [14]) that there exist retractions $p_\beta : Y_\beta \to Z_\beta$ for each $\beta \in \mathcal{B}$. In this way we define

$$Y_\beta^{\min} := p_\beta^{-1}(Z_\beta^{\min}).$$

Then the following result allows us to say something more about the stratification. Let us denote by $\mathfrak{t}_+$ our choice of the positive Weyl chamber in the Lie algebra of some maximal torus of K. Each adjoint orbit intersects $\mathfrak{t}_+$ in a unique point, so $\mathcal{B}$ can be regarded alternatively as a finite set of points in $\mathfrak{t}_+$.

Proposition 5.1.11. *In the previous hypothesis, for each β, we have $S_\beta = GY_\beta^{min}$, and $x \in S_\beta$ if and only if β is the unique closest point to 0 of $\mu(\overline{Gx}) \cap \mathfrak{t}^+$.*

Now, let us denote by $M^{\min}$ the subset of M consisting of points whose paths of steepest descent under the function $f = \|\mu\|^2$ have limit points in $\mu^{-1}(0)$. By 5.1.11 it is a a G-invariant open subset of M. We will need three lemmas.

Lemma 5.1.12. $G = K \exp i\mathfrak{k}$.

Lemma 5.1.13. *If $x \in \mu^{-1}(0)$, then $Gx \cap \mu^{-1}(0) = Kx$.*

Lemma 5.1.14. *Suppose that x and y lie in $\mu^{-1}(0)$ and $x \notin Ky$. Then there exist disjoint G-invariant neighborhoods of x and y in M.*

By following [14], one can show that the following result holds.

Theorem 5.1.15. *Let M be a Kähler manifold acted on by a group G, which is the complexification of a maximal compact subgroup K that preserves the Kähler structure on M. Suppose that a moment map $\mu : M \to \mathfrak{k}$ exists for this action of K, and suppose that the stabilizer of every $x \in \mu^{-1}(0)$ is finite. Then $M^{min} = G\mu^{-1}(0)$, and the natural map $\mu^{-1}(0)/K \to M^{min}/G$ is a homeomorphism.*

5.2 Lagrangian orbits

It is time we begin our analysis on K-homogeneous Lagrangian submanifolds of M, i.e. those Lagrangian submanifolds that are K-invariant, which means those that can be immersed as Lagrangian orbits inside M. We will use the same notations as in section 5.1 and denote by $K^{\mathbb{C}}$ the complexification of a Lie group K.
Let us begin with a strong result obtained in [9], whose proof can be found in [9, pp. 596-597].

Theorem 5.2.1. *Let K be a compact connected group of isometries acting in a Hamiltonian fashion on a compact Kähler manifold M with $h^{1,1}(M) = 1$. Then M admits a K-homogeneous Lagrangian submanifold if and only if $K^{\mathbb{C}}$ has an open Stein orbit in M.*

The definition of Stein orbit is a little analytic, and since we don't need to delve too deep into complex analysis, we will see only the definition and one of its characterization in our hypotheses.

Definition 5.2.2. A complex manifold M is called **Stein manifold** if, by letting $\mathcal{O}(M)$ be the ring of its holomorphic function, the two following conditions hold:

- for every compact subspace $K \subset M$, the subset

$$\left\{ z \in X \;\middle|\; |f(z)| \leq \sup_{w \in K} |f(w)| \; \forall f \in \mathcal{O}(M) \right\} \subset M$$

  is compact as well;

- if $x \neq y$ are two points in M, then there exists $f \in \mathcal{O}(M)$ such that $f(x) \neq f(y)$.

For further topics in the theory if Stein manifolds (and more generally of Stein spaces), [11] is an interesting reference. However it makes use of sheaf theory, so the mathematical language would be different from that of the present thesis. However, there is a more practical characterization available for homogeneous Stein submanifolds (i.e. Stein orbits). In [20] it is proved that

Lemma 5.2.3. *Homogeneous Stein manifolds are those complex homogeneous manifolds for which the stabilizer is the complexification of a compact Lie group.*

From now on, let us suppose that the action allows for at least one Lagrangian orbit, i.e. there exists a open Stein orbit Ω. As before, we let $G := K^{\mathbb{C}}$. Let us define the **set of semistable points**

$$M^{ss} := \{x \in M | \overline{G \cdot x} \cap \mu^{-1}(0) \neq \emptyset\}.$$

We can argue that, whenever the zero level set $\mu^{-1}(0)$ meets the open Stein orbit Ω, then $\Omega = M^{\mathrm{ss}}$. By definition of the set of semistable points, under this hypothesis we have $\Omega \subseteq M^{\mathrm{ss}}$. The opposite inclusion holds either, given that M^{ss} is the smallest open G- invariant subset of M containing $\mu^{-1}(0)$ (cfr. [28, pp. 105–106]). It follows that Ω and M^{ss} do coincide.
Furthermore, a check over the definitions shows that the stratum associated to the minimum critical set of $||\mu||^2$ coincides with the set of semistable points M^{ss}. However, Duistermaat showed (see for instance [17]) that this stratum retracts to the zero set. Until now, we obtained that Ω retracts over $\mu^{-1}(0)$, which implies that they have the same topology.
The natural question now is, what is exactly $\mu^{-1}(0)$? By Kirwan's Lemma 5.1.14, we immediately understand that it is a single orbit. We will need the following.

Lemma 5.2.4. *A orbit Kp through some point $p \in M$ is isotropic if and only if $\mu(p)$ belongs to the center of $\mathfrak{k}$.*

Remark 5.2.5. *We want to specify here that, as always, we are assuming the Lie algebras isomorphism $\mathfrak{k} \simeq \mathfrak{k}^*$. When we say that the moment map $\mu : M \to \mathfrak{k}^*$ is assuming values in the center $Z(\mathfrak{k})$ of the Lie algebra $\mathfrak{k}$, this*

amounts to saying that μ assumes values in $Z(\mathfrak{k})^* \simeq [\tilde{\mathfrak{k}}^*, \tilde{\mathfrak{k}}^*] = \tilde{\mathfrak{k}}^*$, due to the splitting of Lemma 1.2.5.
Also, recall that if the Lie algebra $\mathfrak{k}$ is semisimple, i.e. $\mathfrak{k} = [\mathfrak{k}, \mathfrak{k}]$, then isotropic (and hence Lagrangian) orbits are all sent to the zero by the moment map. As specified above, this implies that there is only one Lagrangian orbit.

Hence we understand that $\mu^{-1}(0)$, which is a single orbit, is moreover isotropic. We finally conclude that it is actually Lagrangian thanks to [9, Lemma 2.1]. We summarize the previous data in the following.

Theorem 5.2.6. *If $\mu^{-1}(0) \cap \Omega$ is not empty, then Ω has $\mu^{-1}(0)$ as a deformation retract, and thus has the same homotopy type of the Lagrangian orbit.*

5.3 Minimal Lagrangian orbits

Finally, one may wonder whether there is a relation between Lagrangian orbits and their *mean curvature vector field H* being zero. We will say that a submanifold is **minimal** whenever its mean curvature vector field vanishes identically. We follow [9].

Proposition 5.3.1. *If K is semisimple and M is KE, the (unique) Lagrangian orbit $\mathcal{O}$ is also minimal.*

The proof exploits the fact that, for any orbit Σ, there is a vector bundle isomorphism

$$T\Sigma^{\perp} \simeq \Lambda^1\Sigma,$$

given by $X \mapsto \iota_X \omega_{|\Sigma}$.
Hence to the mean curvature vector field H of a Lagrangian orbit $\mathcal{O}$ we can associate $\alpha_H \in \Omega^1(\mathcal{O})$. It is clear that one is zero if and only if the other vanishes identically. Also, whenever the metric is KE it has been proven that α_H is closed. Indeed the mathematician Pierre Dazord has proved in [6] that

$$d\alpha_H = \rho_{|\Sigma},$$

where as always ρ is the Ricci form. When M is KE, $\rho = c\omega$ and this shows that α_H is closed, since ω vanishes identically when valued on the tangent bundle of a Lagrangian submanifolds.

Proof. We continue from where we left before and compute, for any $X, Y \in \mathfrak{k}$,

$$0 = d\alpha_H(\widehat{X}, \widehat{Y}) = \widehat{X}\alpha_H(\widehat{Y}) - \widehat{Y}\alpha_H(\widehat{X}) - \alpha([\widehat{X}, \widehat{Y}]).$$

But H is K invariant, since all the ingredients in its definition are, which implies that so is α_H. Then we are left with

$$0 = -\alpha_H([\widehat{X}, \widehat{Y}]) \text{ for every } X, Y \in \mathfrak{k}.$$

We recall that K is semisimple if and only if $[\mathfrak{k}, \mathfrak{k}] = \mathfrak{k}$. Hence we proved that

$$0 = \alpha_H(\widehat{\mathfrak{k}}, \widehat{\mathfrak{k}}) = \alpha_H(\widehat{\mathfrak{k}})$$

and $\alpha_H = 0$. Thus $H = 0$. □

Actually in the KE case one can strengthen this result by mean of the canonical moment map found in Chapter 4, without the further assumption of K being semisimple. Recall that we defined

$$\mu^X = \frac{-1}{4\pi c} \operatorname{div}(J\widehat{X})$$

for any $X \in \mathfrak{k}$.

Proposition 5.3.2. *Let μ be the canonical moment map of a KE manifold (M, J, g), then a Lagrangian orbit $\mathcal{O}$ is minimal if and only if $\mu(\mathcal{O}) = 0$.*

Proof. Since L is Lagrangian, in order to prove that it is minimal, it suffices to show that the mean curvature vector H at some point p of L is orthogonal to $J\widehat{\mathfrak{k}}_p$. Once an orthonormal frame $\{e_i\}$ at p is fixed, we have

$$\begin{aligned} g(H, J\widehat{Y}) &= g(\nabla_{e_i} e_i, J\widehat{Y}) \\ &= e_i g(e_i, J\widehat{Y}) - g(e_i, \nabla_{e_i} J\widehat{Y}) \\ &= -g(e_i, \nabla_{e_i} J\widehat{Y}) \\ &= -\frac{1}{2}\operatorname{div} J\widehat{Y} \\ &= 2\pi c\mu(Y) = 0. \end{aligned}$$

Notice that we have used the expression $H = \nabla_{e_i} e_i$ on the orbit (see [24]). Also, the second equality follows from the compatibility of the Levi-Civita connection ∇ with the metric g. We remark that e_i and $J\widehat{Y}$ are orthogonal, which gives us $g(e_i, J\widehat{Y}) = 0$. □

By mean of this last result and section 4.2, we get the following corollaries.

Corollary 5.3.3. *Let K be a compact connected group of isometries acting in a Hamiltonian fashion on a compact KE manifold M. Then M admits at most one minimal Lagrangian K-orbit.*

Corollary 5.3.4. *Under the same hypotheses of Theorem 1.1, assuming further that $G = K^{\mathbb{C}}$ is simply connected, M is KE, and the isotropy subgroup G_q at a point q of the Stein orbit has finite connected components, we get that M admits a unique Lagrangian K-orbit which turns out to be minimal.*

So far we investigated the material contained in [9]. Let us now go a bit further.
Suppose that (M, J, g) is a compact Kähler manifold and assume furthermore that K is compact and semisimple, and that it acts holomorphically and isometrically in the usual fashion. We need to weave some facts together.
Let us suppose first that Lagrangian orbits constitute a stratum inside M. Being the orbit $\mathcal{O}$ Lagrangian and the Lie group K semisimple, one has that

$$\mu(\mathcal{O}) = 0,$$

where μ is the unique and existent (cfr. section 2.4) moment map associated to the action.
From Kirwan's Lemma 5.1.14 it follows that in $\mu^{-1}(0)$ sits only one orbit, and we have seen that it is Lagrangian.
The mathematician Hsiang shows in [13] that there is at least one minimal submanifold among orbits of each type. Since there is only one Lagrangian orbit, we have proven that

Lemma 5.3.5. *Whenever Lagrangian orbits constitute a stratum and the Lie group K is semisimple, there is only one Lagrangian orbit and it is minimal.*

5.4 Final remarks and perspectives

We conclude this thesis with a brief remark and some perspectives for further investigation.
In the setting of Chapter 4, we were able to compute the formula (4.13) for the moment map of a KRS along a vector field X:

$$\mu^X = \left(\frac{-\operatorname{div} + df}{4\pi} \right) (J\widehat{X}).$$

Whenever the acting Lie group K is semisimple, we observed that any Lagrangian orbit $\mathcal{O}$ is sent to the zero by the moment map μ, assuming that it exists. By applying Lemma 5.3.5, if we assume that Lagrangian orbits constitute a stratum we can conclude that there exists only one Lagrangian orbit and that it is minimal. The proof of Proposition 5.3.2 gives us that $\operatorname{div}(J\widehat{X}) = 0$ for any $X \in \mathfrak{k}$. Hence, on the Lagrangian orbit it holds

$$0 = \mu^X = \left(\frac{-\operatorname{div} + df}{4\pi}\right)(J\widehat{X}) = \frac{df}{4\pi}(J\widehat{X}).$$

Being $\widehat{X}$ K-invariant, it also happens that

$$df(\widehat{X}) = 0,$$

for any $X \in \mathfrak{k}$. But then we have proved that

Lemma 5.4.1. *In the previous setting,*

$$\mathcal{O} \subseteq \operatorname{Crit}(f),$$

i.e. the unique (minimal) Lagrangian orbit is contained inside the set of critical points of f.

We can even infer more information about the map f involved in the KRS equation of the manifold M.

Remark 5.4.2. *Recall that in (4.16) we were able to show that f is a multiple of the moment map μ along $J\nabla f$. As such, f is a non-degenerate Morse function on M in the sense of Bott by Proposition 5.1.4.*

We leave here two interesting points that might be explored as continuation of the themes touched in the present thesis.

- What happens whenever K is not semisimple? The starting point would be the analysis of the most extreme example of non-semisimple Lie group, i.e. abelian Lie groups.
- What relation exists between $||\mu||^2$ and f when K is again semisimple?

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Fede

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